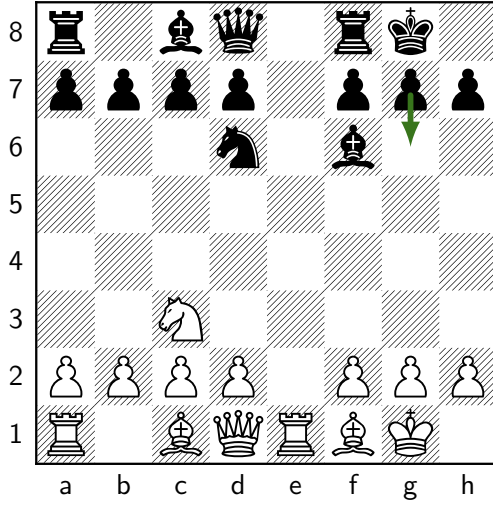


Melanie-BOTS (corpus: CCRL engines) — match game 5

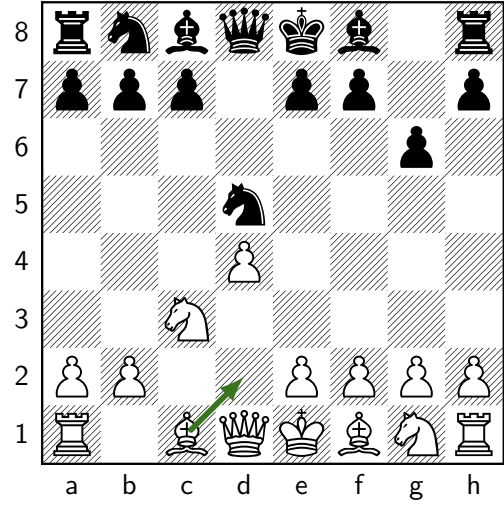
Melanie (Deep_Vision)

Abstract

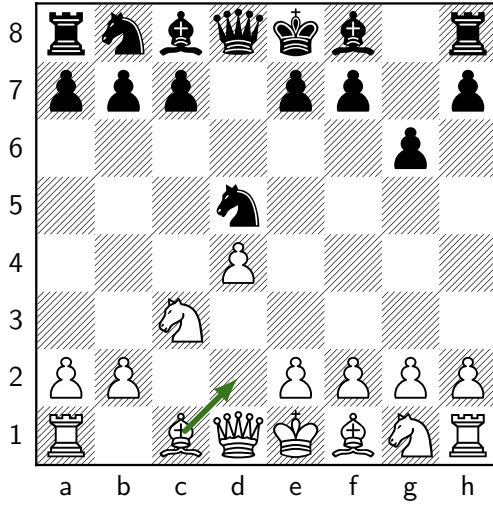
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



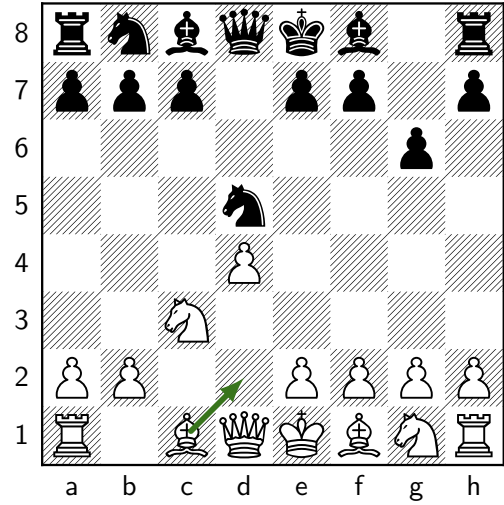
(a)



(b)

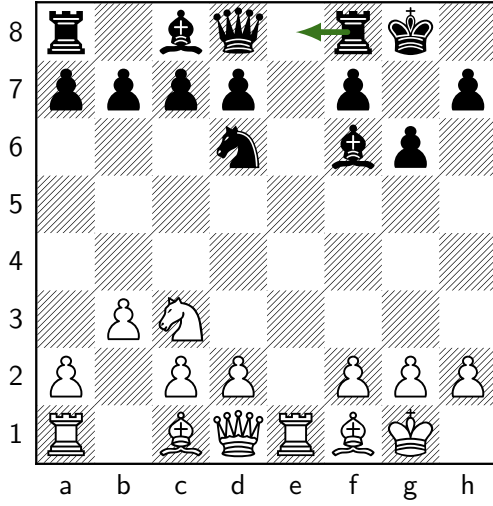


(c)

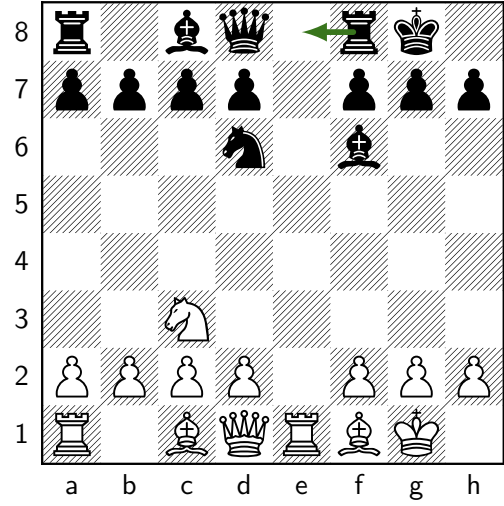


(d)

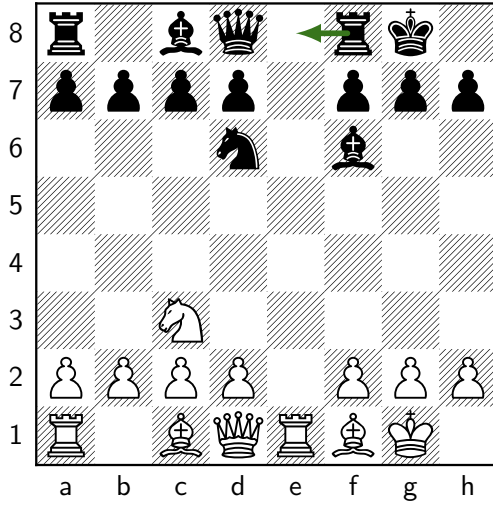
Figure 1: **Move 10, Black: g6**. (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#1216 (ply 8) [structural distance $d=64$, QAP obj=19]: the master played Bd2; under μ our Pg2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Rf8 as their Qd1, our ph2 as their Ng1; transferred move g7g6 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#1217 (ply 8) [structural distance $d=64$, QAP obj=19]: the master played Bd2; under μ our Pg2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Rf8 as their Qd1, our ph2 as their Ng1; transferred move g7g6 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3300 (ply 8) [structural distance $d=64$, QAP obj=19]: the master played Bd2; under μ our Pg2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Rf8 as their Qd1, our ph2 as their Ng1; transferred move g7g6 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to g6 (summed QAP objective 260, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



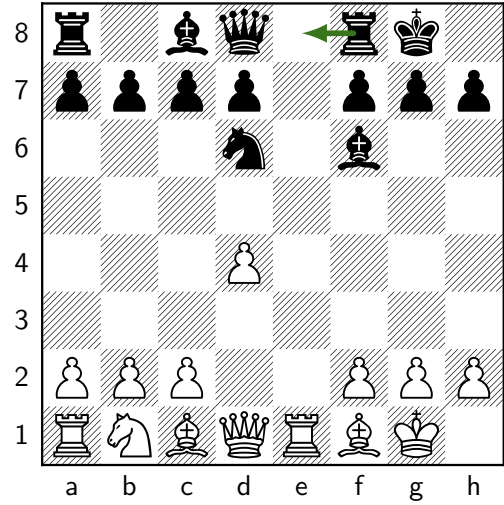
(a)



(b)

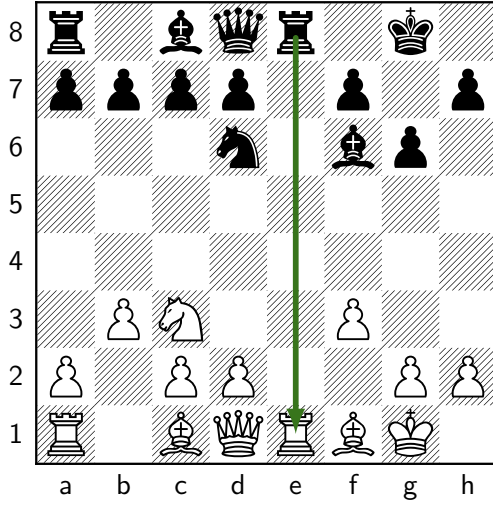


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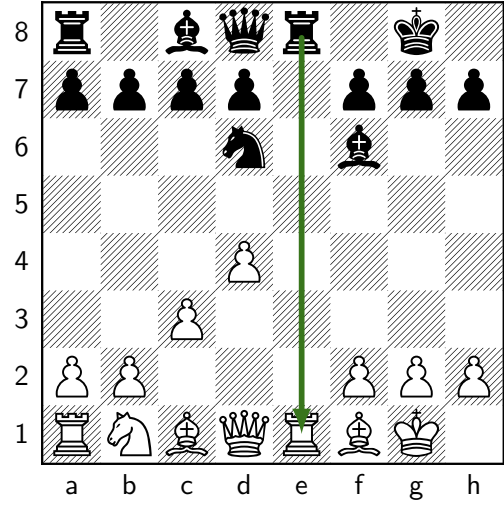


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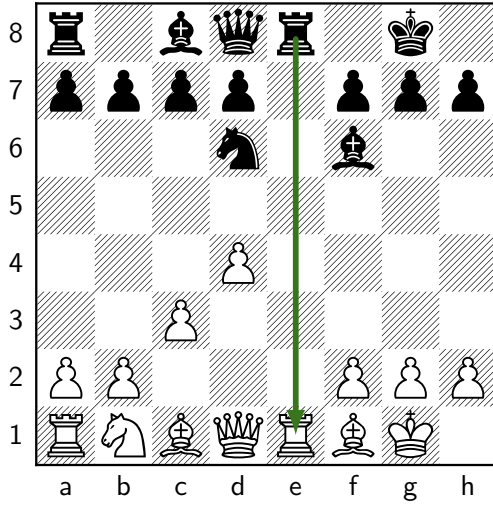
Figure 2: **Move 11, Black: Re8.** (a) Our position; Melanie plays **Re8** (**f8e8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1677` (ply 19) [structural distance $d=21$, QAP obj=32]: the master played **Re8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pc2* as their *Ra8*, our *re1* as their *Bc8*; transferred move **f8e8** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3182` (ply 19) [structural distance $d=21$, QAP obj=32]: the master played **Re8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pc2* as their *Ra8*, our *re1* as their *Bc8*; transferred move **f8e8** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1075` (ply 19) [structural distance $d=54$, QAP obj=16]: the master played **Re8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Pa7* as their *Ra8*, our *Ph7* as their *Bc8*; transferred move **f8e8** (**endorses this move**). Synthesis: 12 retrieved analogs transferred to **Re8** (summed QAP objective 217, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



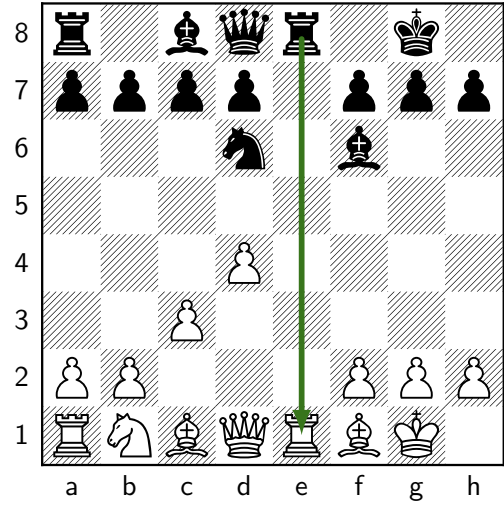
(a)



(b)

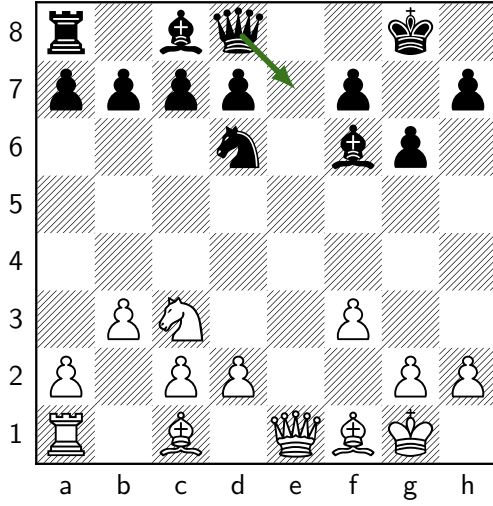


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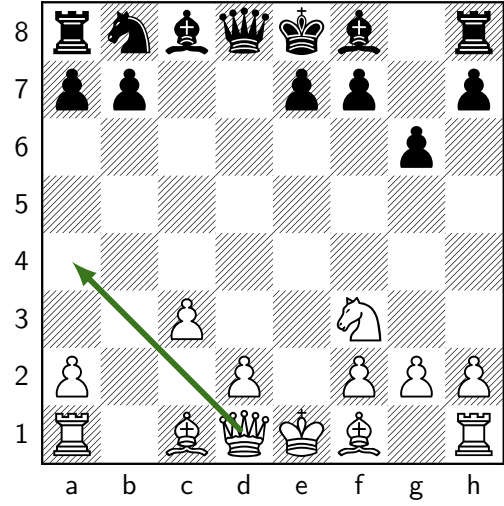


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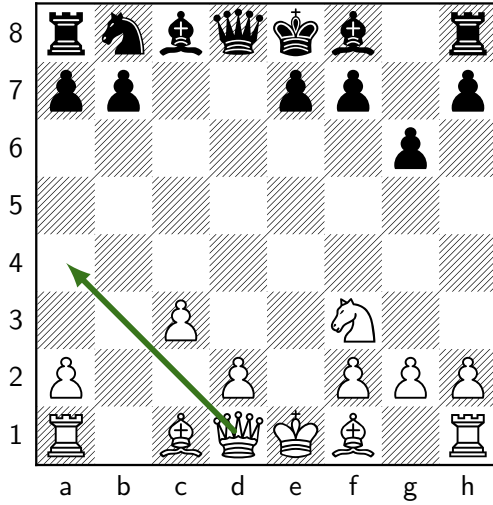
Figure 3: **Move 12, Black: Rxe1.** (a) Our position; Melanie plays Rxe1 (e8e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#784` (ply 21) [structural distance $d=46$, QAP obj=57]: the master played Rxe1; under μ our Re1 plays the role of that game’s Re1; cross-type roles: our bc1 as their Pg7, our Re8 as their Bf6; transferred move e8e1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#701` (ply 21) [structural distance $d=46$, QAP obj=57]: the master played Rxe1; under μ our Re1 plays the role of that game’s Re1; cross-type roles: our bc1 as their Pg7, our Re8 as their Bf6; transferred move e8e1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#741` (ply 21) [structural distance $d=46$, QAP obj=57]: the master played Rxe1; under μ our Re1 plays the role of that game’s Re1; cross-type roles: our bc1 as their Pg7, our Re8 as their Bf6; transferred move e8e1 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rxe1 (summed QAP objective 419, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



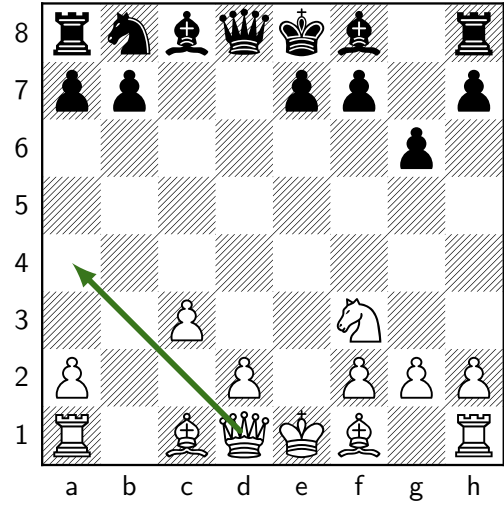
(a)



(b)

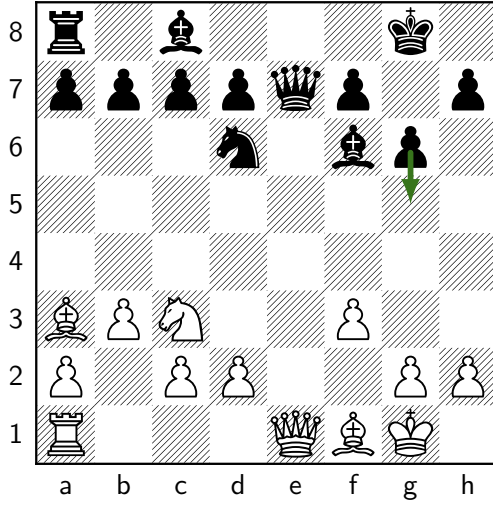


(c)

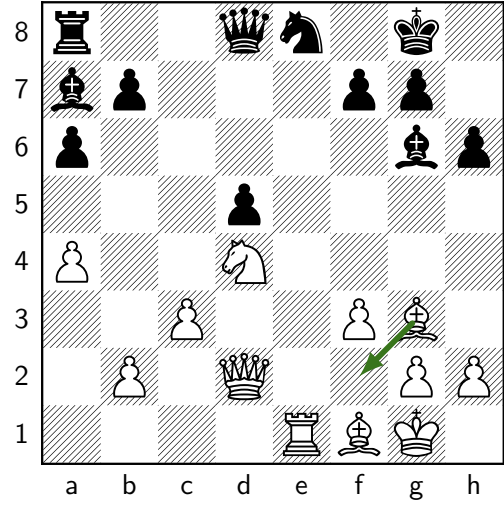


(d)

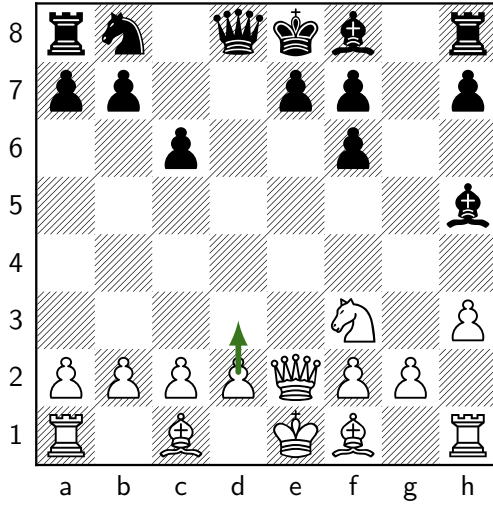
Figure 4: **Move 13, Black: Qe7.** (a) Our position; Melanie plays Qe7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3050` (ply 14) [structural distance $d=70$, QAP obj=5]: the master played Qa4+; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf7 as their Ra1, our qe1 as their Bc1; transferred move d8e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3051` (ply 14) [structural distance $d=70$, QAP obj=5]: the master played Qa4+; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf7 as their Ra1, our qe1 as their Bc1; transferred move d8e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#751` (ply 14) [structural distance $d=70$, QAP obj=5]: the master played Qa4+; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf7 as their Ra1, our qe1 as their Bc1; transferred move d8e7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Qe7 (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



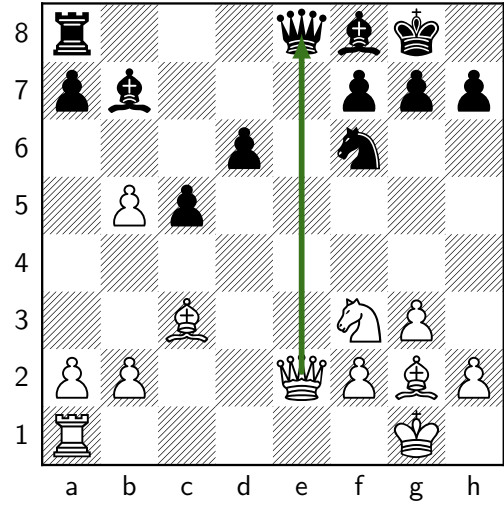
(a)



(b)

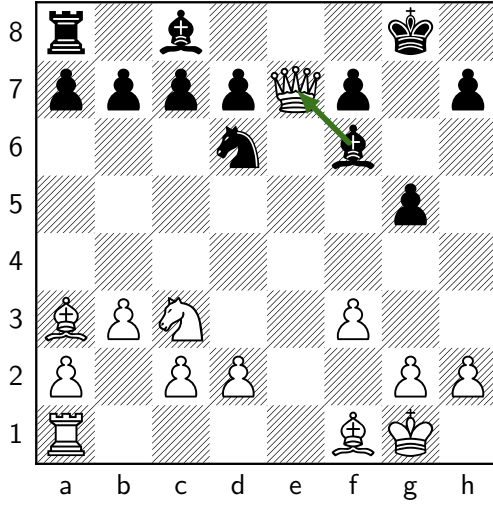


(c)

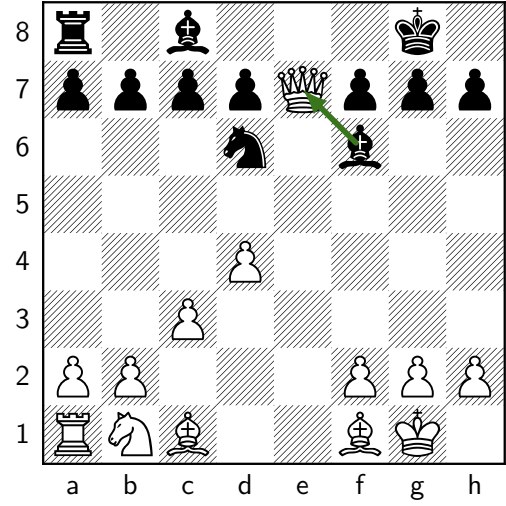


(d)

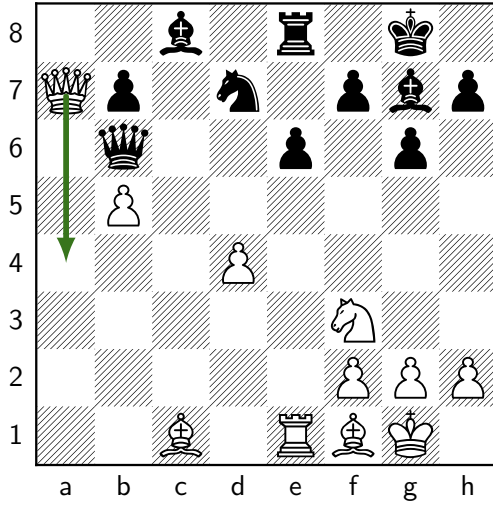
Figure 5: **Move 14, Black: g5.** (a) Our position; Melanie plays g5 (g6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3163` (ply 40) [structural distance $d=74$, QAP obj=49]: the master played Bf2; under μ our Pg3 plays the role of that game’s Bg3 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Re1, our Bc8 as their Pb2; transferred move g6g5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1082` (ply 14) [structural distance $d=86$, QAP obj=31]: the master played d3; under μ our Pg3 plays the role of that game’s Pd2; cross-type roles: our Bf6 as their Ra1, our Ph7 as their Bc1; transferred move g6g5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3699` (ply 38) [structural distance $d=82$, QAP obj=67]: the master played Qxe8; under μ our Qe2 plays the role of that game’s Qe2; cross-type roles: our Pf7 as their Ra1, our Bc8 as their Pa2; transferred move e7e1. Synthesis: 2 retrieved analogs transferred to g5 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



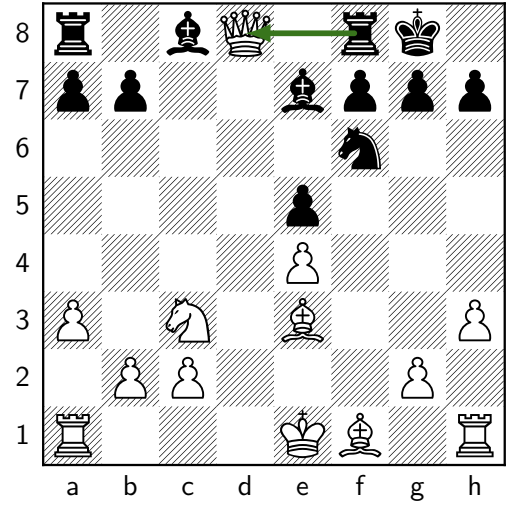
(a)



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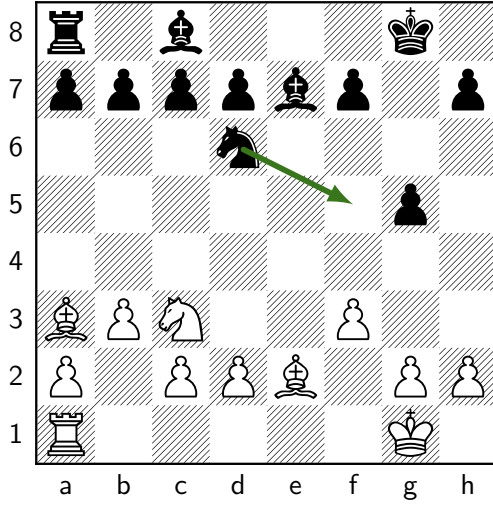


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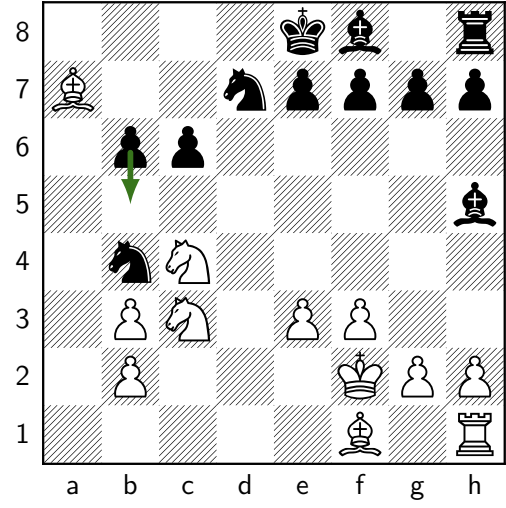


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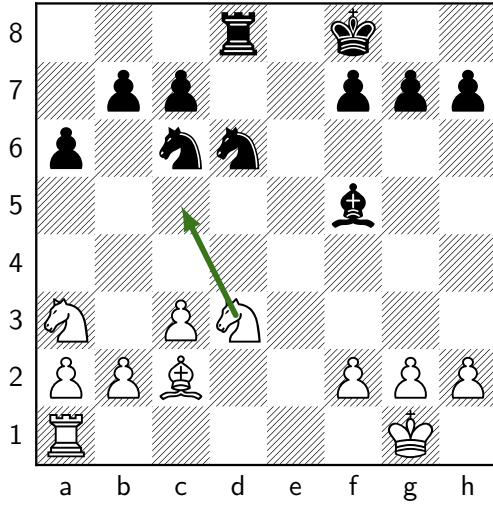
Figure 6: **Move 15, Black: Bxe7 (f6e7).** (a) Our position; Melanie plays Bxe7 (f6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#742` (ply 25) [structural distance $d=43$, QAP obj=69]: the master played Bxe7; under μ our Bf3 plays the role of that game’s Bf3; cross-type roles: our Pc7 as their Ra8, our nc3 as their pd4; transferred move f6e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3245` (ply 42) [structural distance $d=113$, QAP obj=67]: the master played Qa4; under μ our Bf3 plays the role of that game’s Qa7 (a **cross-type** correspondence); cross-type roles: our Pg5 as their Bc1, our ba3 as their Re1; transferred move f6e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2825` (ply 23) [structural distance $d=113$, QAP obj=48]: the master played Rxd8; under μ our Bf3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our Nd6 as their Bc8; transferred move f6e7 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Bxe7 (summed QAP objective 469, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



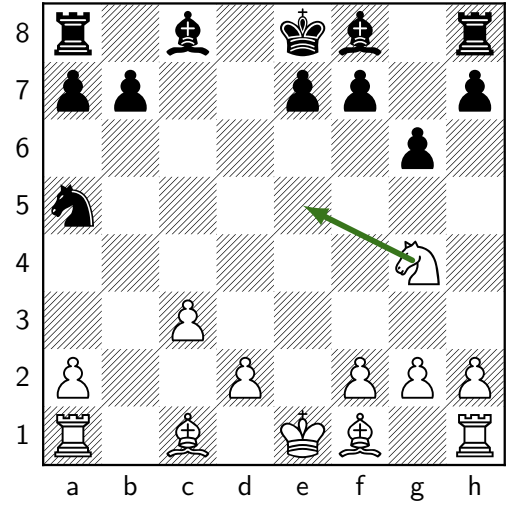
(a)



(b)



(c)



(d)

Figure 7: **Move 16, Black: Nf5.** (a) Our position; Melanie plays Nf5 (d6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#513 (ply 35) [structural distance $d=71$, QAP obj=27]: the master played b5; under μ our Nd3 plays the role of that game’s Pb3 (a **cross-type** correspondence); cross-type roles: our Pd7 as their Bf8, our Be7 as their Nd7; transferred move d6f5 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#663 (ply 36) [structural distance $d=72$, QAP obj=20]: the master played Nc5; under μ our Nd3 plays the role of that game’s Nd3; cross-type roles: our pd2 as their Ra1, our Bc8 as their Pf2; transferred move d6f5 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3051 (ply 22) [structural distance $d=76$, QAP obj=17]: the master played Ne5; under μ our Nd3 plays the role of that game’s Ng4; cross-type roles: our Pg5 as their Bf1, our pd2 as their Rh1; transferred move d6f5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Nf5 (summed QAP objective 150, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

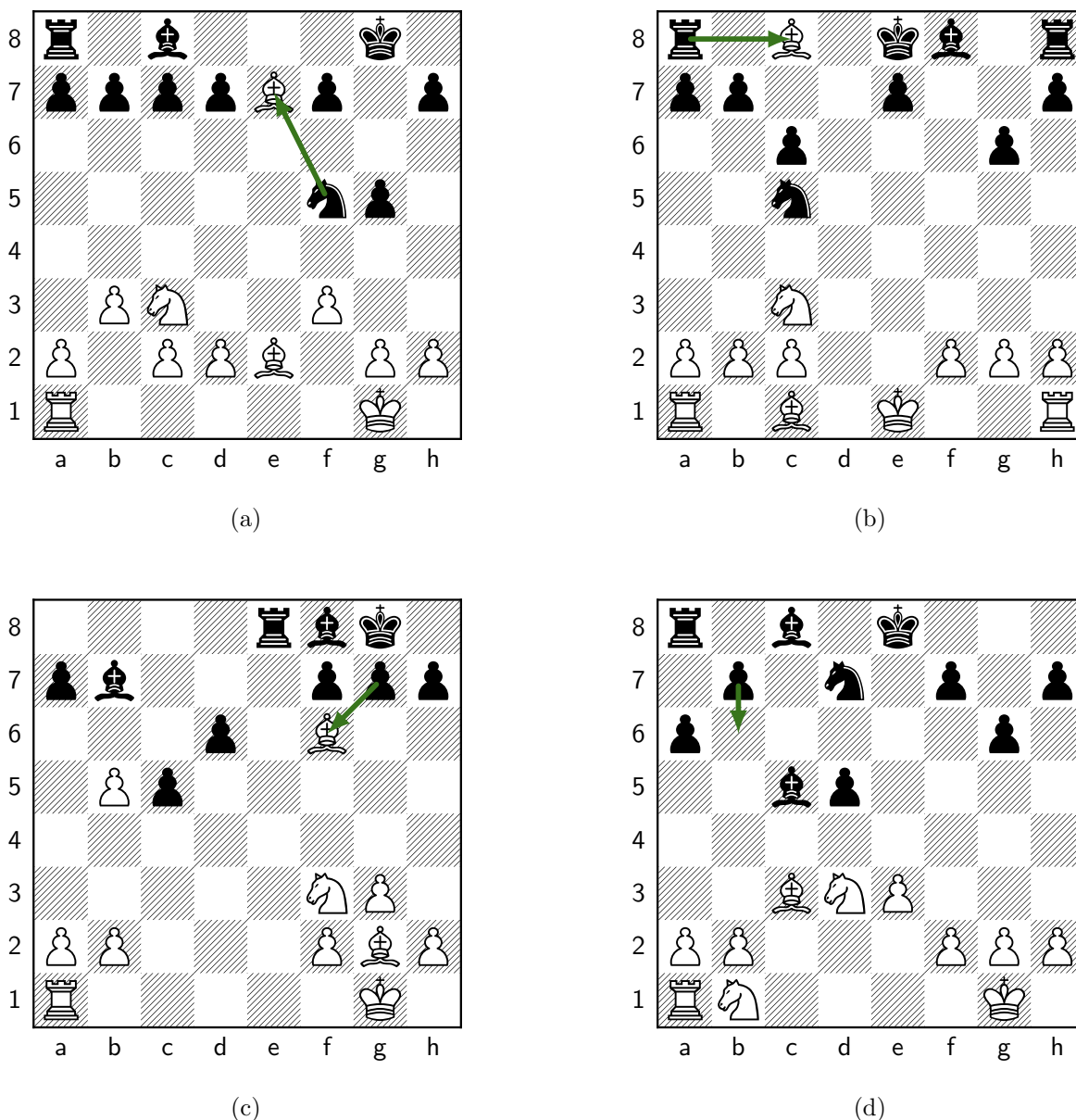
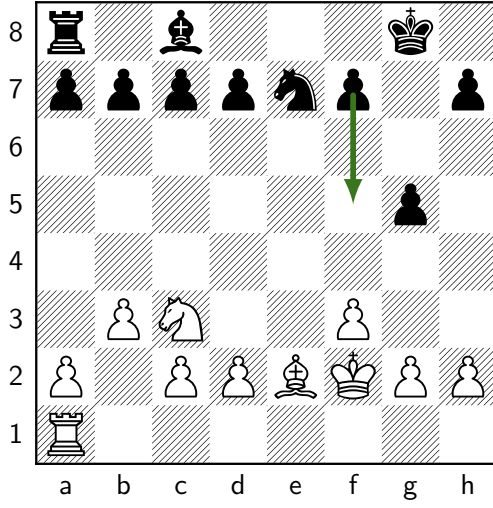
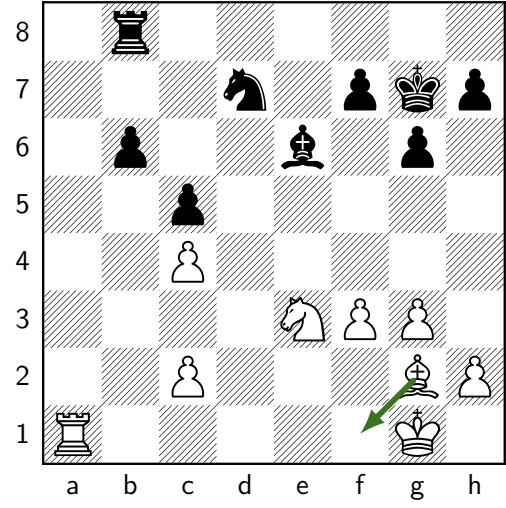


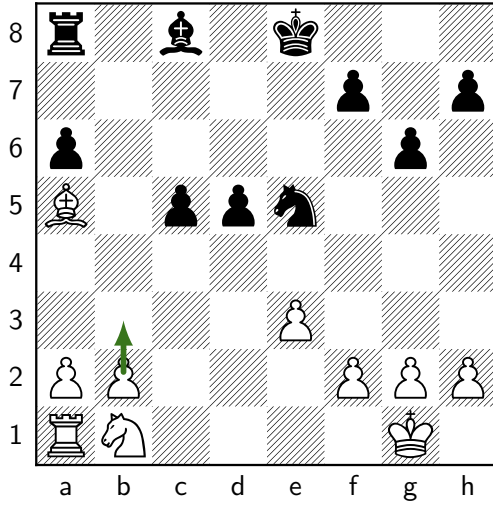
Figure 8: **Move 17, Black: Nxe7.** (a) Our position; Melanie plays Nxe7 (f5e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#987` (ply 23) [structural distance $d=63$, QAP obj=44]: the master played Rxc8; under μ our Nf4 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Nc5, our pg2 as their nc3; transferred move f5e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3699` (ply 41) [structural distance $d=52$, QAP obj=14]: the master played gxf6; under μ our Nf4 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our nc3 as their Re8, our Pf7 as their Bf8; transferred move f5e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#176` (ply 35) [structural distance $d=65$, QAP obj=45]: the master played b6; under μ our Pb2 plays the role of that game’s Pb2; cross-type roles: our Pf7 as their Nd7, our be2 as their Pa6; transferred move b7b6. Synthesis: 2 retrieved analogs transferred to Nxe7 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



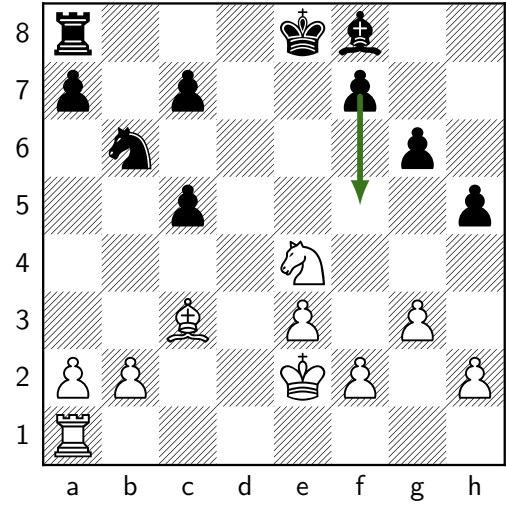
(a)



(b)



(c)



(d)

Figure 9: **Move 18, Black: f5 (f7f5).** (a) Our position; Melanie plays f5 (f7f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2113` (ply 50) [structural distance $d=65$, QAP obj=4]: the master played Bf1; under μ our Pf2 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Ne7 as their Ra1, our Ra8 as their Pc2; transferred move f7f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#176` (ply 40) [structural distance $d=62$, QAP obj=3]: the master played b3; under μ our Pf2 plays the role of that game’s Pb2; cross-type roles: our Bc8 as their Ra1, our Pc7 as their Nb1; transferred move f7f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2628` (ply 41) [structural distance $d=60$, QAP obj=3]: the master played f5; under μ our Pf2 plays the role of that game’s Pf2; cross-type roles: our Ne7 as their Ra8, our pb3 as their Nb6; transferred move f7f5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to f5 (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

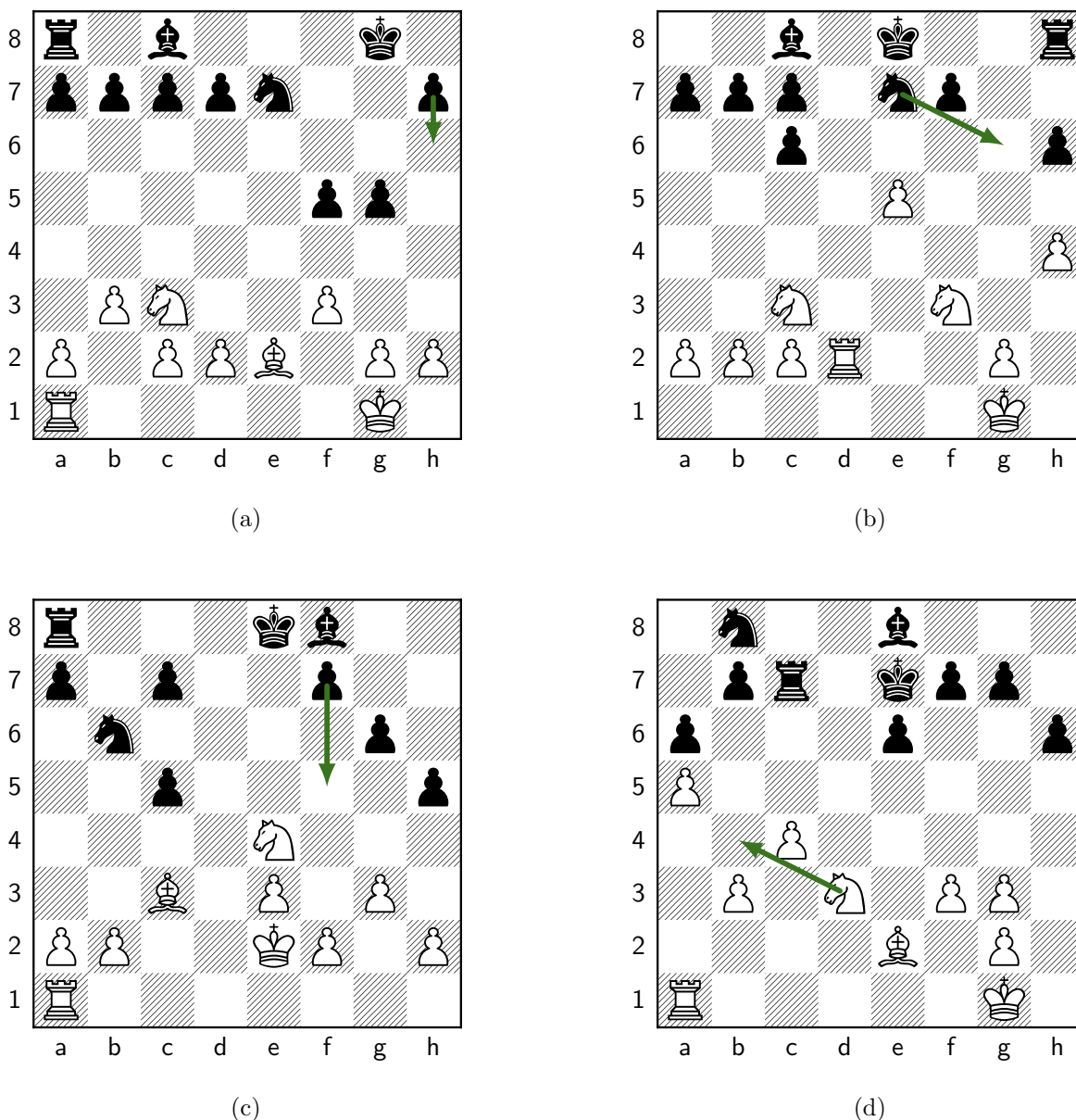
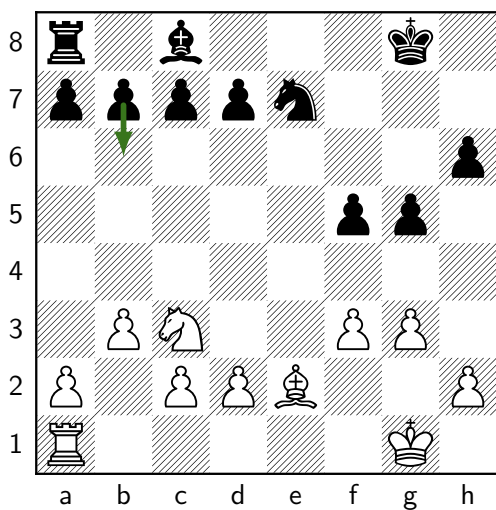
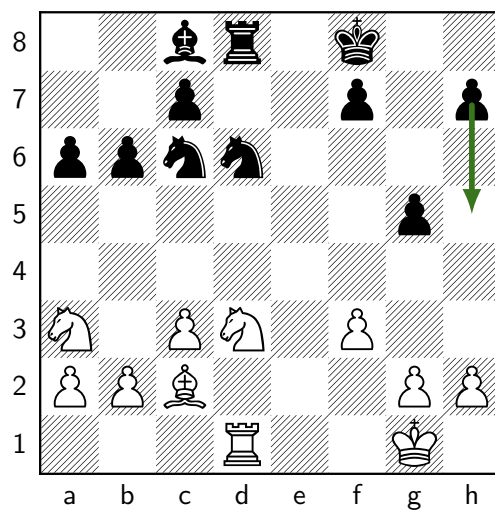


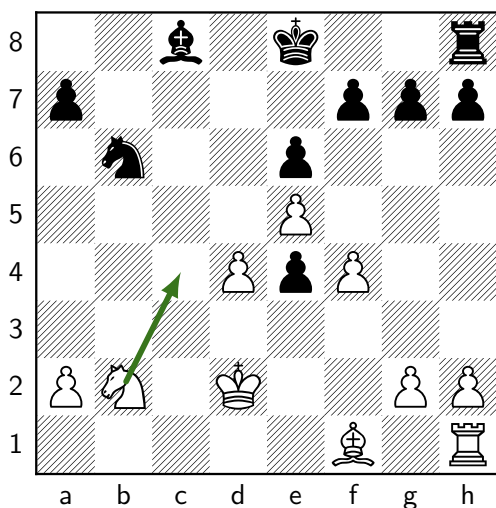
Figure 10: **Move 19, Black: h6**. (a) Our position; Melanie plays h6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2525` (ply 41) [structural distance $d=66$, QAP obj=4]: the master played Ng6; under μ our *Ph2* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *Pg5* as their *Bc8*, our *Ne7* as their *Rh8*; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2628` (ply 41) [structural distance $d=64$, QAP obj=3]: the master played f5; under μ our *Ph2* plays the role of that game’s *Pf2*; cross-type roles: our *Ne7* as their *Ra8*, our *pd2* as their *Bf8*; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2909` (ply 52) [structural distance $d=61$, QAP obj=3]: the master played Nb4; under μ our *Ph2* plays the role of that game’s *Nd3* (a **cross-type** correspondence); cross-type roles: our *Ne7* as their *Ra1*, our *Ra8* as their *Be2*; transferred move h7h6 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to h6 (summed QAP objective 22, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



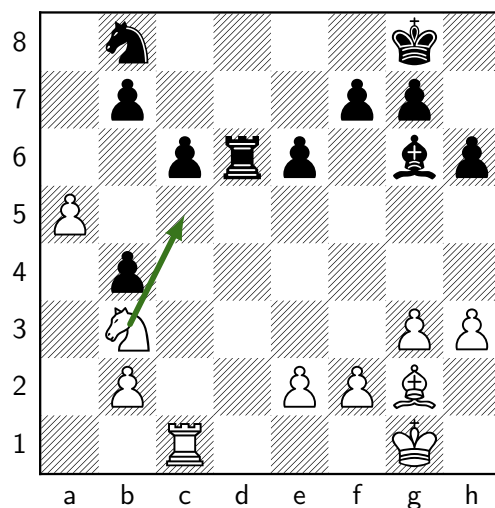
(a)



(b)



(c)



(d)

Figure 11: **Move 20, Black: b6**. (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#663` (ply 43) [structural distance $d=68$, QAP obj=5]: the master played h5; under μ our *Pb2* plays the role of that game’s *Ph2*; cross-type roles: our *pf3* as their *Bc8*, our *Ne7* as their *Rd8*; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1750` (ply 44) [structural distance $d=57$, QAP obj=4]: the master played Nc4; under μ our *Pb2* plays the role of that game’s *Nb2* (a **cross-type** correspondence); cross-type roles: our *pb3* as their *Bf1*, our *nc3* as their *Pg2*; transferred move b7b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2185` (ply 54) [structural distance $d=63$, QAP obj=4]: the master played Nc5; under μ our *Pb2* plays the role of that game’s *Nb3* (a **cross-type** correspondence); cross-type roles: our *Ne7* as their *Rc1*, our *Ra8* as their *Pb2*; transferred move b7b6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to b6 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

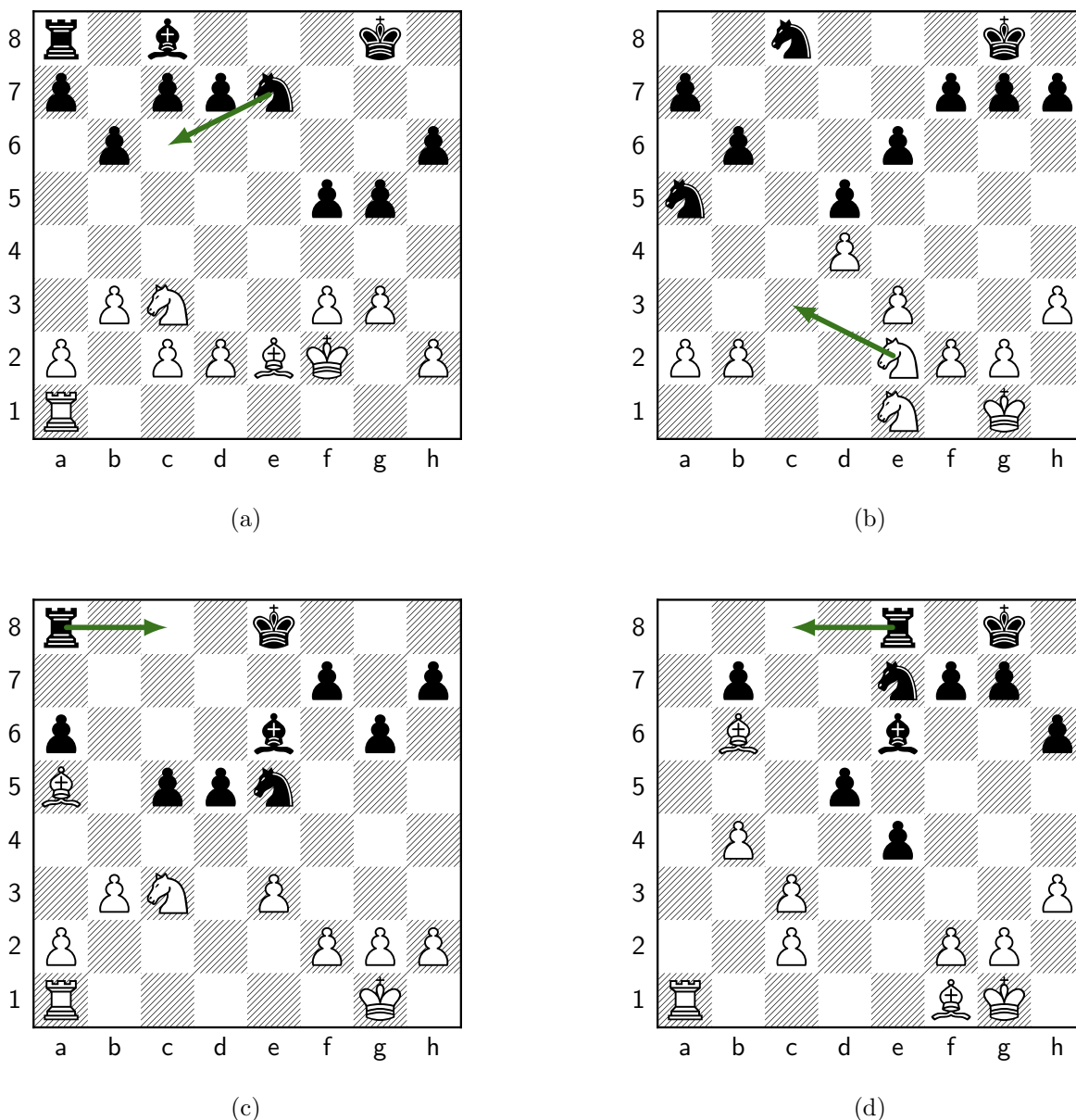
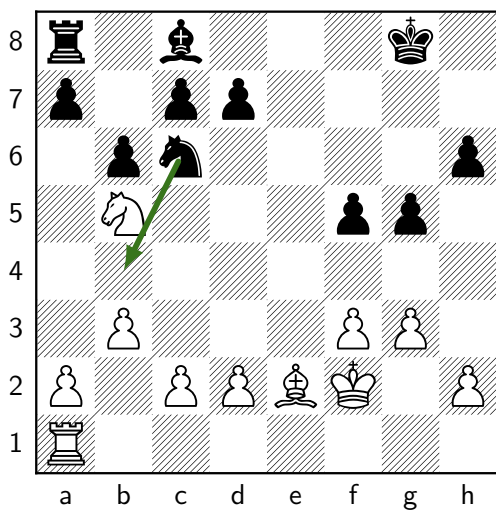
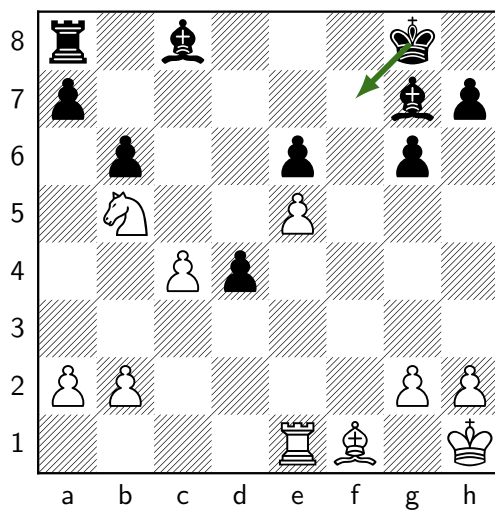


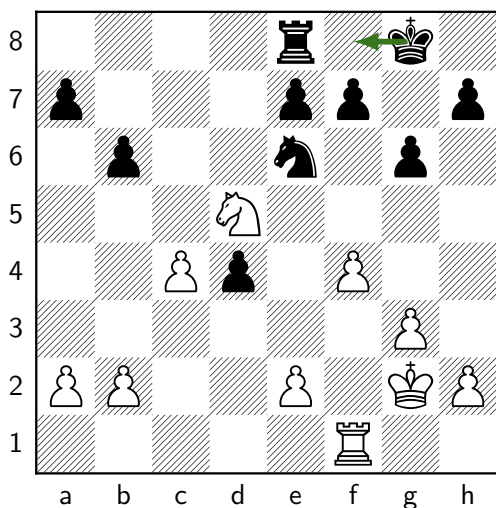
Figure 12: **Move 21, Black: Nc6.** (a) Our position; Melanie plays Nc6 (e7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#483 (ply 48) [structural distance $d=58$, QAP obj=5]: the master played Nc3; under μ our Ne2 plays the role of that game’s Ne2; cross-type roles: our Ra8 as their Ne1, our be2 as their Pf2; transferred move e7c6 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#176 (ply 43) [structural distance $d=56$, QAP obj=5]: the master played Rc8; under μ our Ne2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pf7, our Bc8 as their Ph7; transferred move e7c6 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#43 (ply 49) [structural distance $d=66$, QAP obj=5]: the master played Rc8; under μ our Ne2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ne7, our Ra8 as their Pg7; transferred move e7c6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nc6 (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



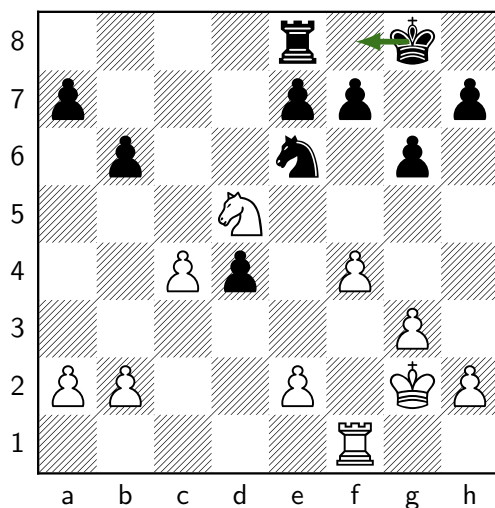
(a)



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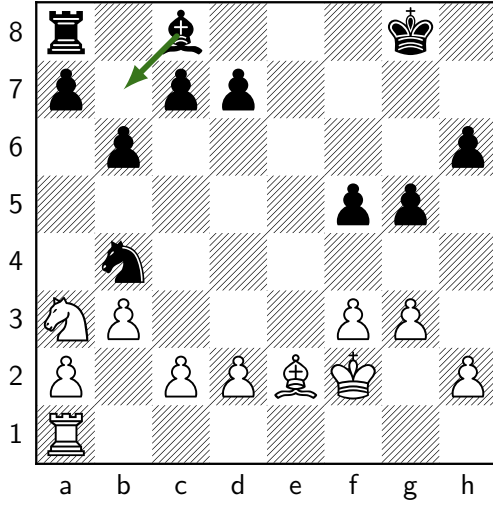


(c)

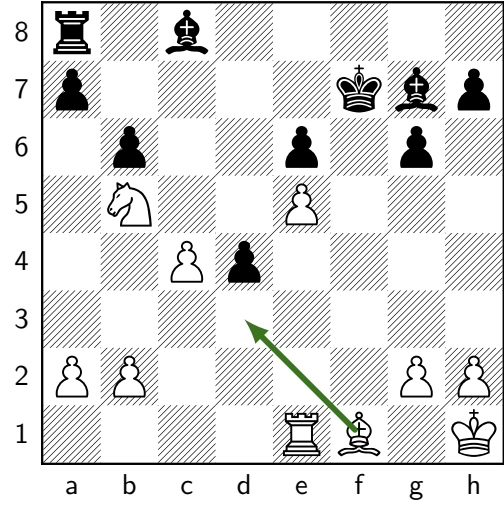


(d)

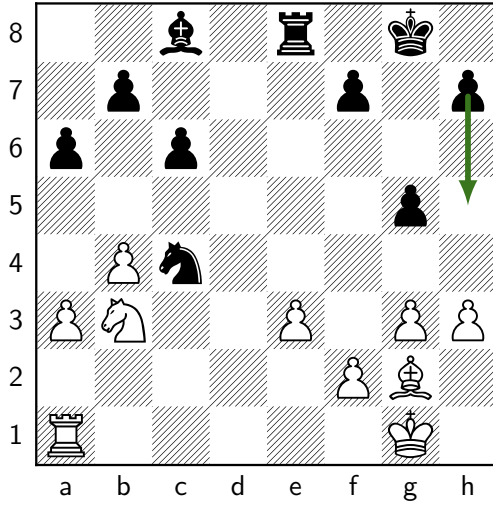
Figure 13: **Move 22, Black: Nb4.** (a) Our position; Melanie plays Nb4 (c6b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1023` (ply 41) [structural distance $d=56$, QAP obj=58]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Ph6 as their Ra8, our Pg5 as their Bc8; transferred move g8g7. (c) Retrieved from `ccr1_top_p0000#1711` (ply 43) [structural distance $d=45$, QAP obj=55]: the master played Kf8; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Bc8 as their Pa7, our Nc6 as their Pg6; transferred move g8h7. (d) Retrieved from `ccr1_top_p0001#3336` (ply 43) [structural distance $d=45$, QAP obj=55]: the master played Kf8; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Bc8 as their Pa7, our Nc6 as their Pg6; transferred move g8h7. Synthesis: no retrieved analog transferred to Nb4 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



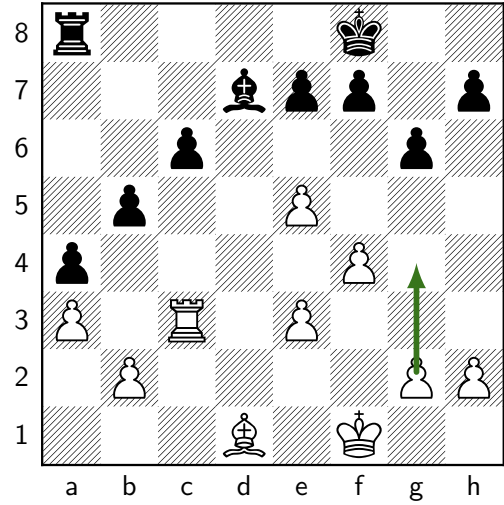
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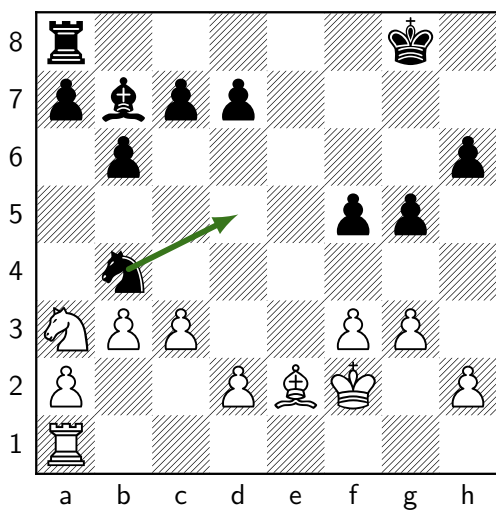


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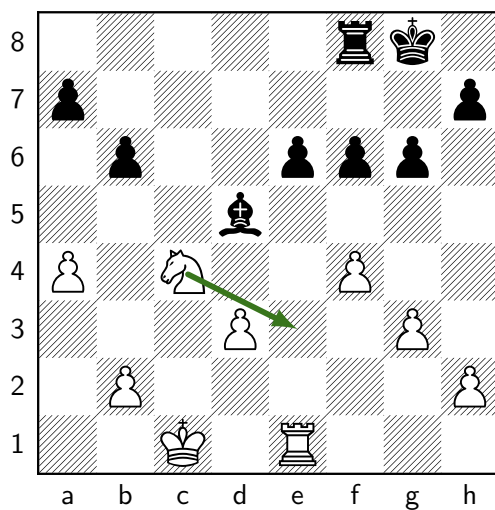


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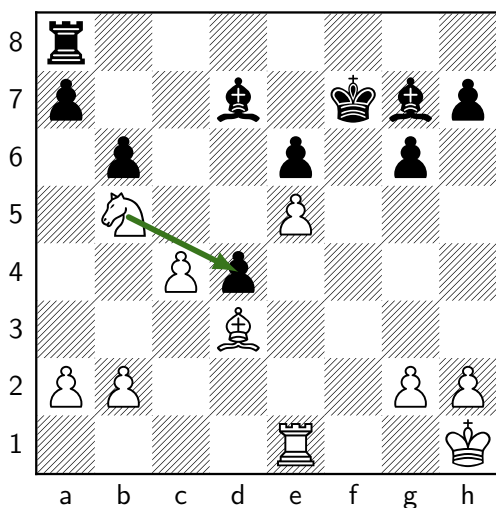
Figure 14: **Move 23, Black: Bb7**. (a) Our position; Melanie plays Bb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0001#1023` (ply 42) [structural distance $d=58$, QAP obj=60]: the master played Bd3; under μ our Bc1 plays the role of that game’s Bf1; cross-type roles: our na3 as their Pb2, our be2 as their pe6; transferred move c8b7 (**endorses this move**). (c) Retrieved from `ccr1.top_p0001#1201` (ply 47) [structural distance $d=56$, QAP obj=50]: the master played h5; under μ our Bc1 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our pf3 as their Bc8, our Nb4 as their Re8; transferred move c8b7 (**endorses this move**). (d) Retrieved from `ccr1.top_p0000#3820` (ply 52) [structural distance $d=56$, QAP obj=24]: the master played g4; under μ our Bc1 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Nb4 as their Bd1, our Ra8 as their Ph2; transferred move c8b7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bb7 (summed QAP objective 135, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



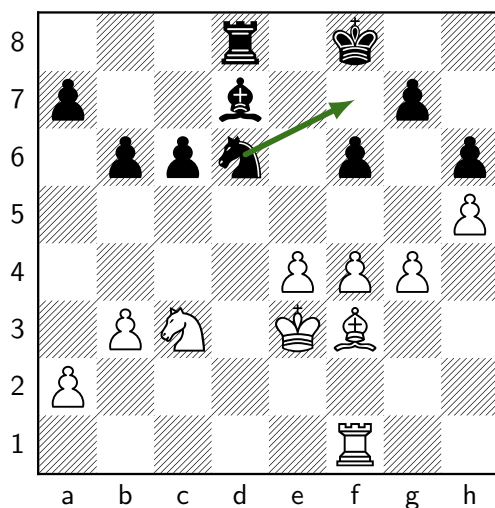
(a)



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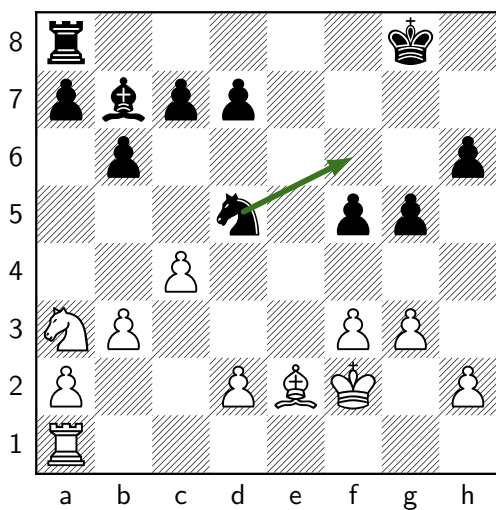


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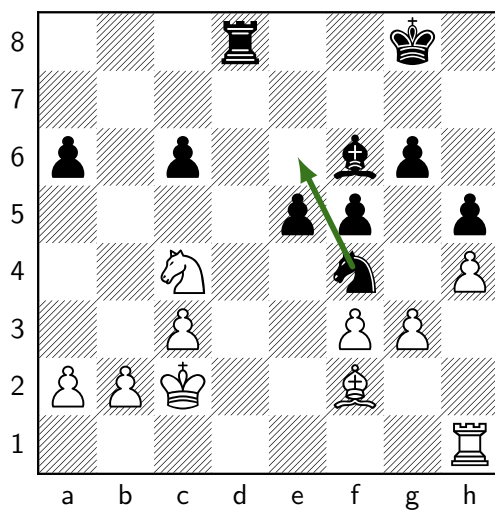


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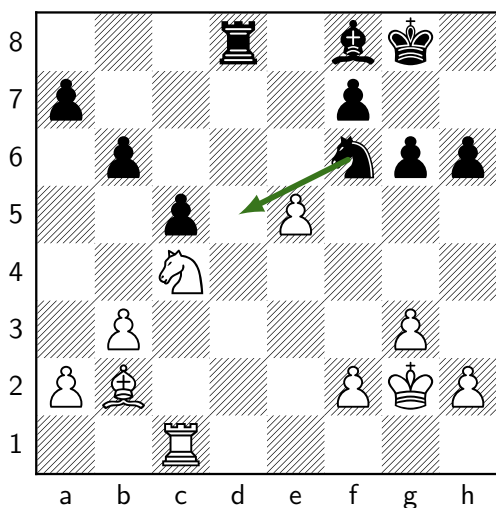
Figure 15: **Move 24, Black: Nd5.** (a) Our position; Melanie plays Nd5 (b4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2100` (ply 48) [structural distance $d=55$, QAP obj=39]: the master played Ne3; under μ our Nb5 plays the role of that game’s Nc4; cross-type roles: our Bb7 as their Re1, our Ra8 as their Pb2; transferred move b4d5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1023` (ply 44) [structural distance $d=55$, QAP obj=33]: the master played Nxd4; under μ our Nb5 plays the role of that game’s Nb5; cross-type roles: our na3 as their Pb2, our Pg5 as their Bd3; transferred move b4d5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3355` (ply 49) [structural distance $d=49$, QAP obj=6]: the master played Nf7; under μ our Nb5 plays the role of that game’s Nd3; cross-type roles: our na3 as their Bd7, our Bb7 as their ph5; transferred move b4d5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nd5 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



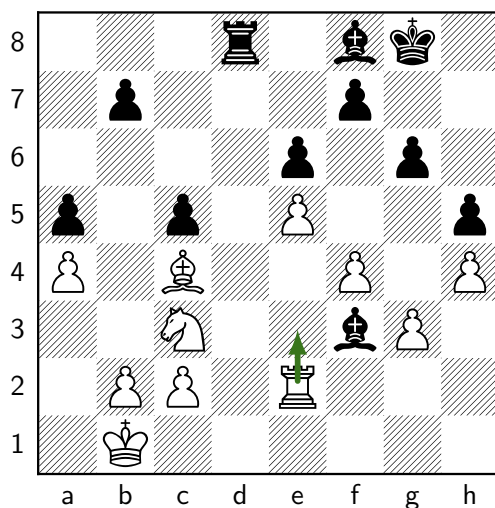
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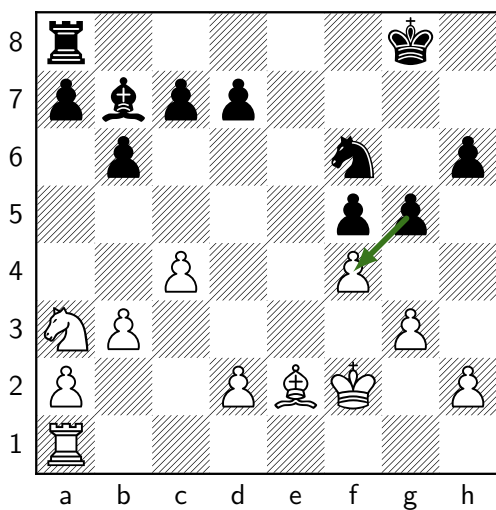


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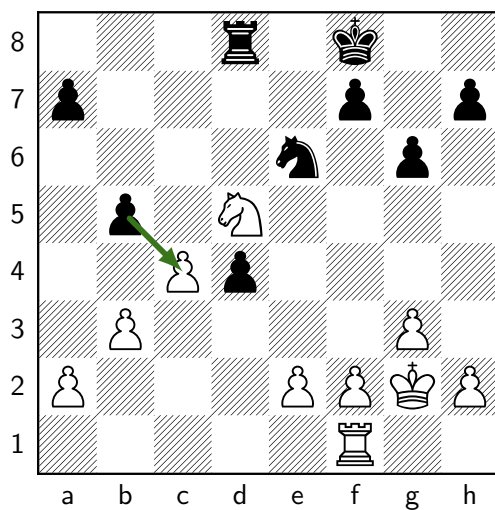


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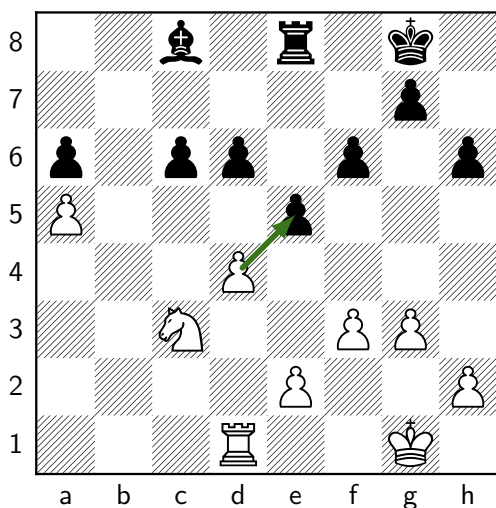
Figure 16: **Move 25, Black: Nf6**. (a) Our position; Melanie plays Nf6 (d5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#106` (ply 51) [structural distance $d=42$, QAP obj=26]: the master played Ne6; under μ our Nd4 plays the role of that game’s Nf5; cross-type roles: our ra1 as their nc4, our na3 as their bf2; transferred move d5f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1232` (ply 51) [structural distance $d=66$, QAP obj=24]: the master played Nd5; under μ our Nd4 plays the role of that game’s Nf3; cross-type roles: our Bb7 as their Rd8, our Pb6 as their Bf8; transferred move d5f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1933` (ply 54) [structural distance $d=68$, QAP obj=17]: the master played Re3; under μ our Nd4 plays the role of that game’s Re2 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pb2, our Bb7 as their Nc3; transferred move d5f6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nf6 (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



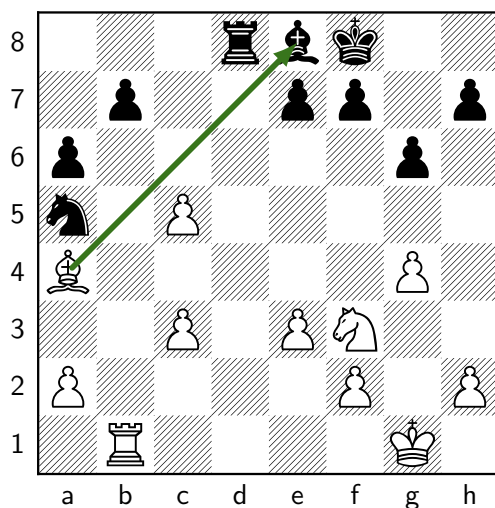
(a)



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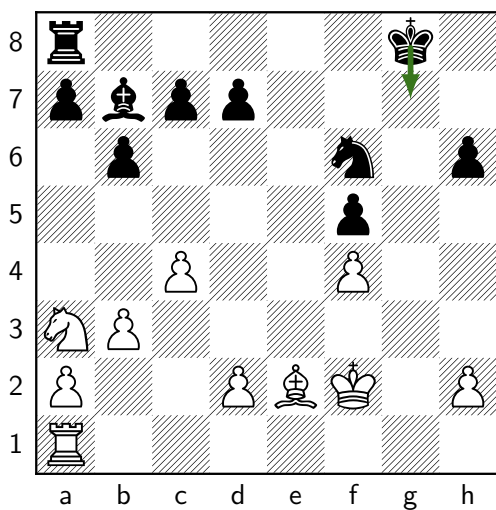


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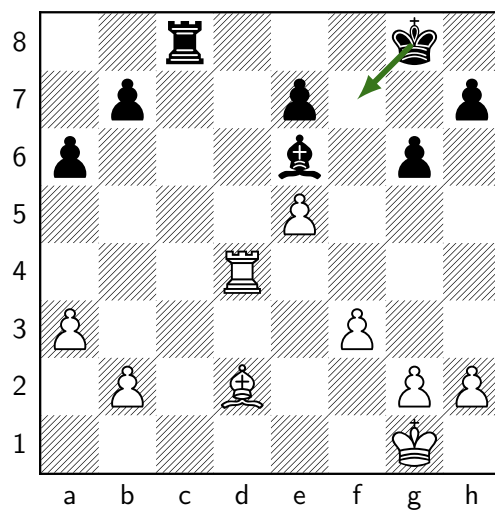


(d)

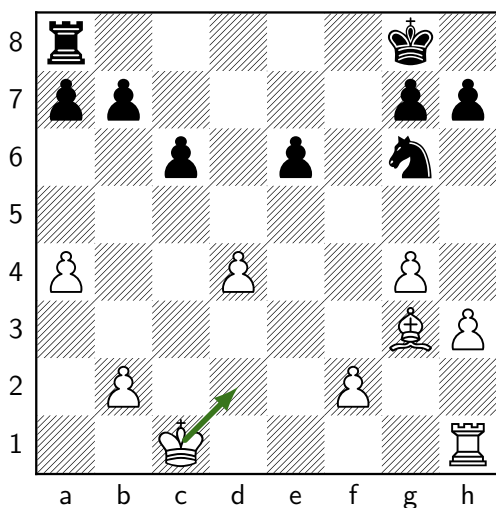
Figure 17: **Move 26, Black: gxf4**. (a) Our position; Melanie plays **gxf4** (**g5f4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1194` (ply 47) [structural distance $d=56$, QAP obj=51]: the master played **bxc4**; under μ our $Pg4$ plays the role of that game’s $Pb4$; cross-type roles: our $Pg5$ as their $Rd8$, our $Ra8$ as their $Pa7$; transferred move **g5f4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3568` (ply 44) [structural distance $d=43$, QAP obj=45]: the master played **dxe5**; under μ our $Pg4$ plays the role of that game’s $Pd4$; cross-type roles: our $Nf6$ as their $Rd1$, our $Bb7$ as their $Pe2$; transferred move **g5f4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3048` (ply 44) [structural distance $d=56$, QAP obj=33]: the master played **Bxe8**; under μ our $Pg4$ plays the role of that game’s $Ba4$ (a **cross-type** correspondence); cross-type roles: our $be2$ as their $Pf2$, our $Pb6$ as their $na5$; transferred move **g5f4** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **gxf4** (summed QAP objective 160, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



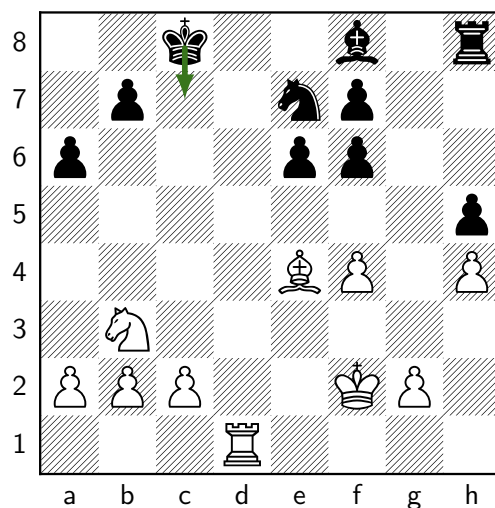
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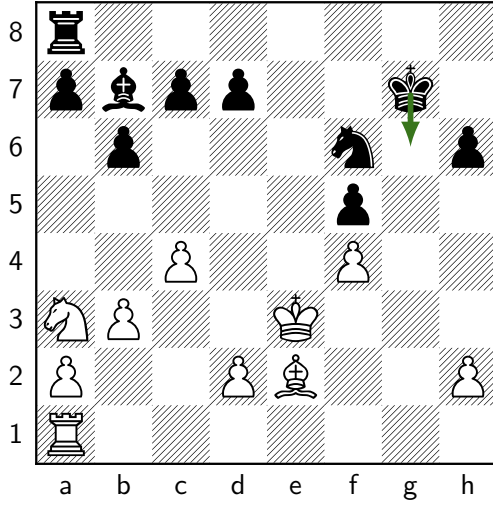


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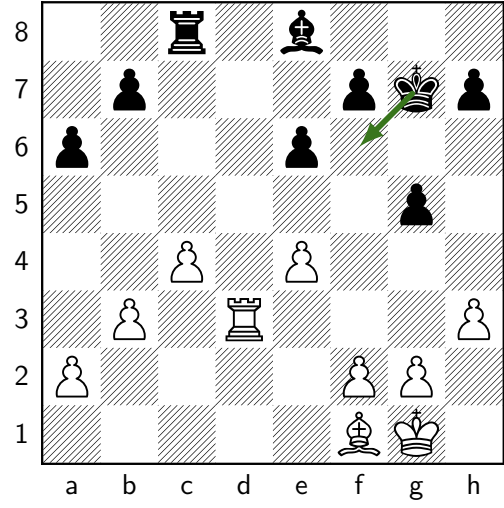


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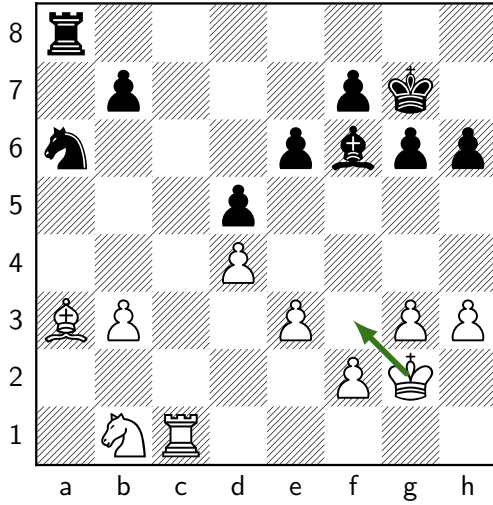
Figure 18: **Move 27, Black: Kg7.** (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2692` (ply 61) [structural distance $d=46$, QAP obj=5]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Nf6 as their Be6, our be2 as their rd4; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1705` (ply 46) [structural distance $d=41$, QAP obj=4]: the master played Kd2; under μ our Kg1 plays the role of that game’s Kc1; cross-type roles: our Nf6 as their Rh1, our Ra8 as their Pb2; transferred move g8g7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2818` (ply 43) [structural distance $d=45$, QAP obj=4]: the master played Kc7; under μ our Kg1 plays the role of that game’s Kc1; cross-type roles: our Pc7 as their Rh8, our na3 as their Pa6; transferred move g8g7 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kg7 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



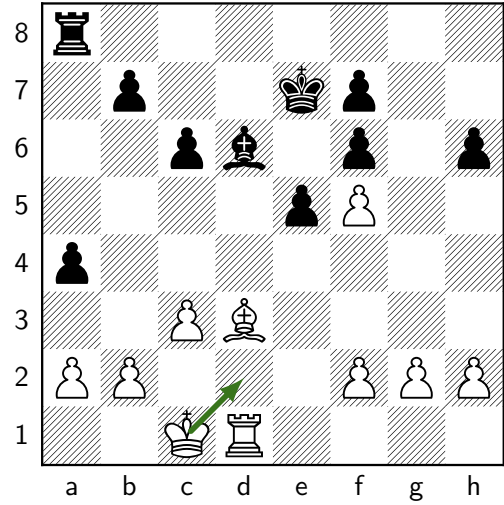
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(b)

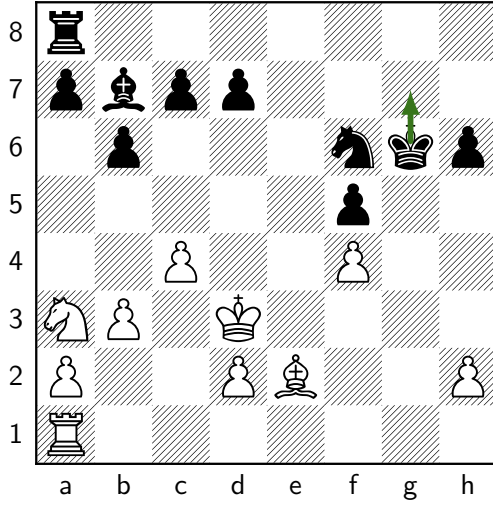


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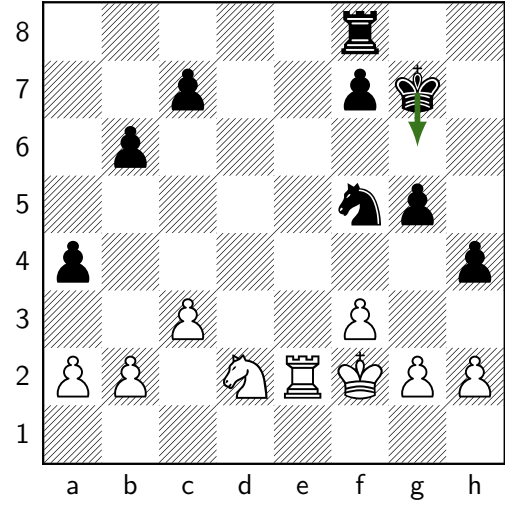


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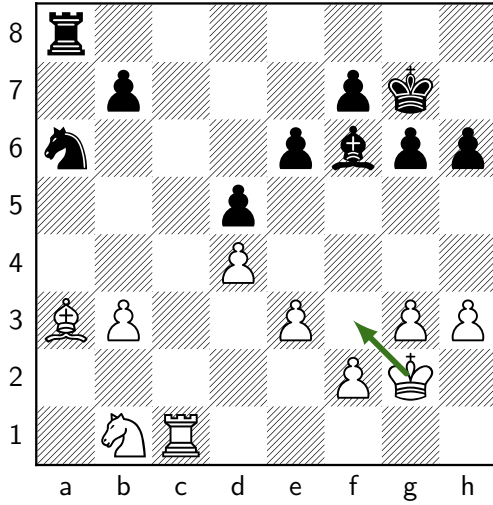
Figure 19: **Move 28, Black: Kg6.** (a) Our position; Melanie plays Kg6 ($g7g6$). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1072` (ply 57) [structural distance $d=38$, QAP obj=4]: the master played Kf6; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Bb7 as their Rc8, our Ra8 as their Be8; transferred move $g7g6$ (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1776` (ply 56) [structural distance $d=32$, QAP obj=3]: the master played Kf3; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Pd7 as their Nb1, our Ph6 as their Rc1; transferred move $g7g6$ (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2037` (ply 42) [structural distance $d=35$, QAP obj=3]: the master played Kd2; under μ our Kg2 plays the role of that game’s Kc1; cross-type roles: our pd2 as their Rd1, our Ra8 as their Pf2; transferred move $g7g6$ (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kg6 (summed QAP objective 22, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



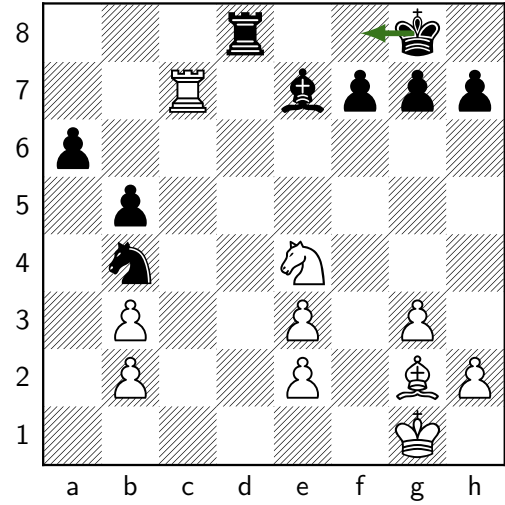
(a)



(b)



(c)



(d)

Figure 20: **Move 29, Black: Kg7.** (a) Our position; Melanie plays Kg7 (**g6g7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#663` (ply 65) [structural distance $d=42$, QAP obj=5]: the master played Kg6; under μ our Kg3 plays the role of that game’s Kg2; cross-type roles: our Pf5 as their Rf8, our Nf6 as their Pf7; transferred move g6g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1776` (ply 56) [structural distance $d=30$, QAP obj=4]: the master played Kf3; under μ our Kg3 plays the role of that game’s Kg2; cross-type roles: our Ra8 as their Nb1, our Pb6 as their Rc1; transferred move g6g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#896` (ply 45) [structural distance $d=43$, QAP obj=3]: the master played Kf8; under μ our Kg3 plays the role of that game’s Kg1; cross-type roles: our pc4 as their Rd8, our pf4 as their rc7; transferred move g6g7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kg7 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

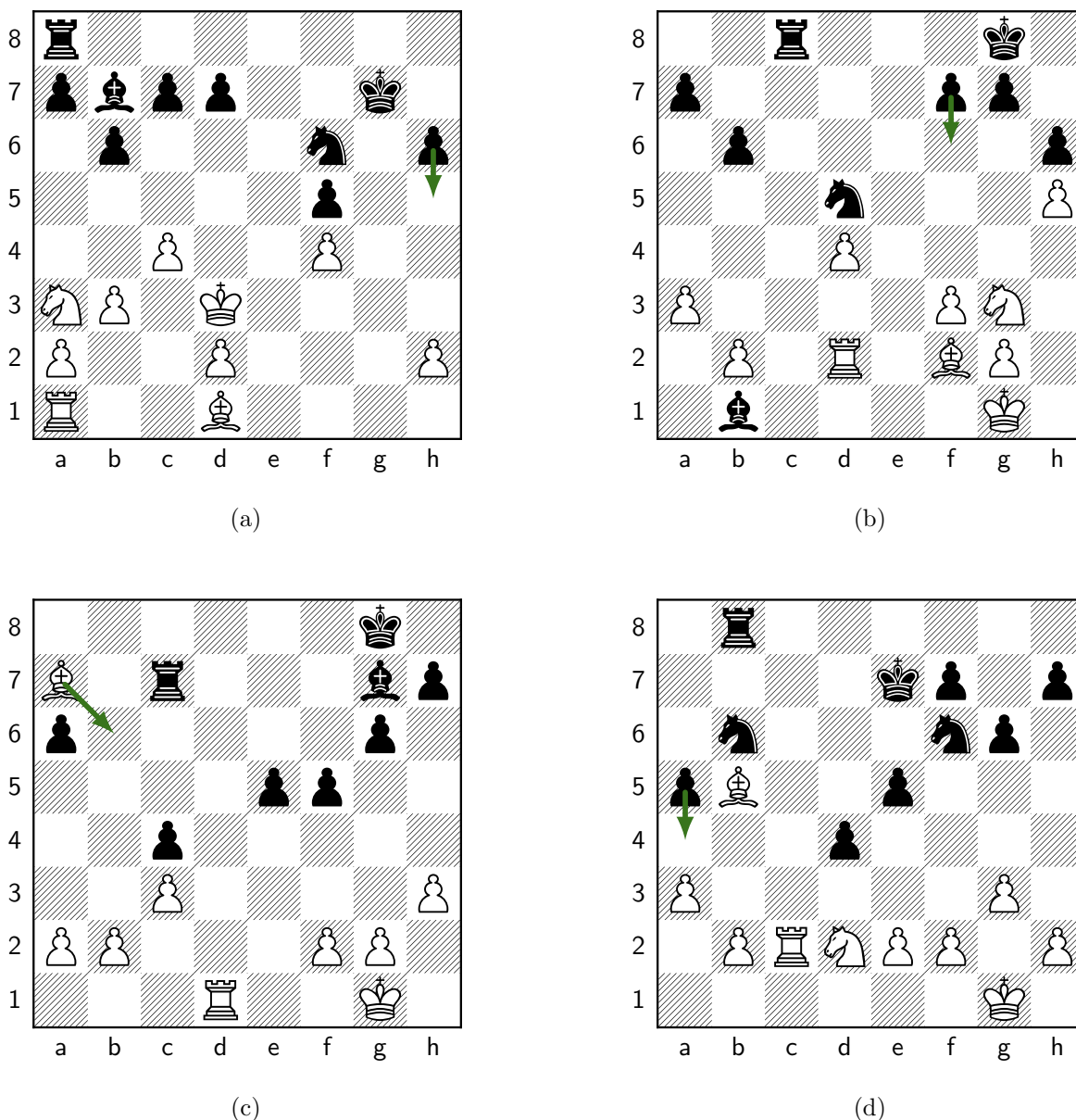
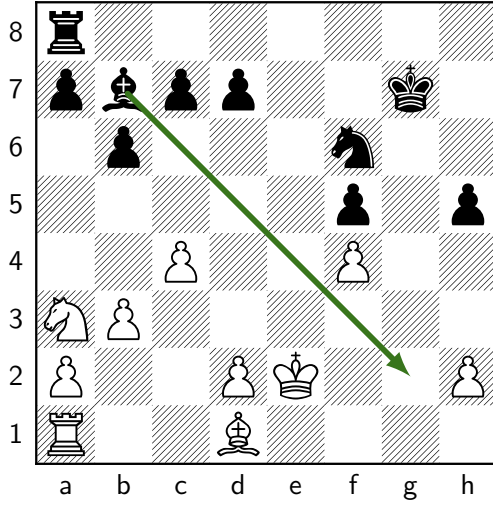
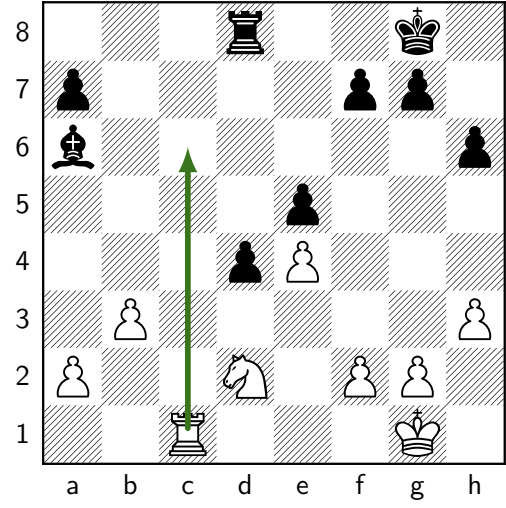


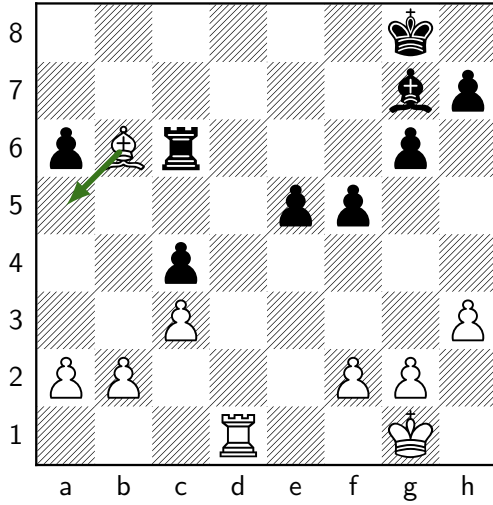
Figure 21: **Move 30, Black: h5**. (a) Our position; Melanie plays h5 (h6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1311` (ply 53) [structural distance $d=38$, QAP obj=5]: the master played f6; under μ our *Ph3* plays the role of that game’s *Pf2*; cross-type roles: our *ra1* as their *pd4*, our *bd1* as their *ng3*; transferred move h6h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3425` (ply 52) [structural distance $d=41$, QAP obj=4]: the master played Bb6; under μ our *Ph3* plays the role of that game’s *Ba7* (a **cross-type** correspondence); cross-type roles: our *Bb7* as their *Rd1*, our *Ra8* as their *Pa2*; transferred move h6h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#302` (ply 51) [structural distance $d=35$, QAP obj=3]: the master played a4; under μ our *Ph3* plays the role of that game’s *Pa4*; cross-type roles: our *Pf5* as their *Nb6*, our *na3* as their *Pg6*; transferred move h6h5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to h5 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



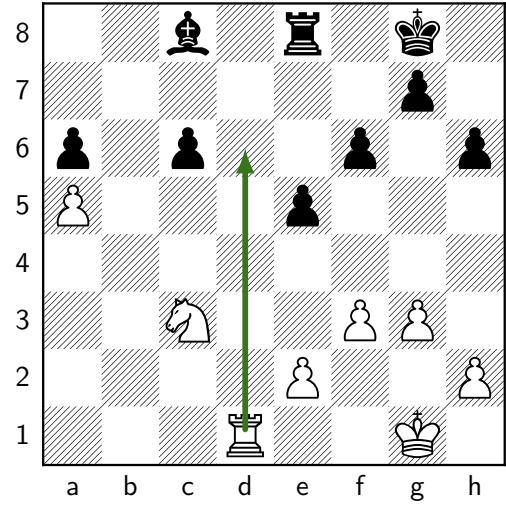
(a)



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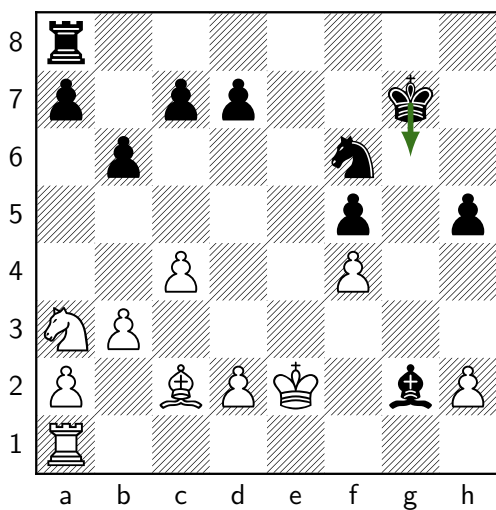


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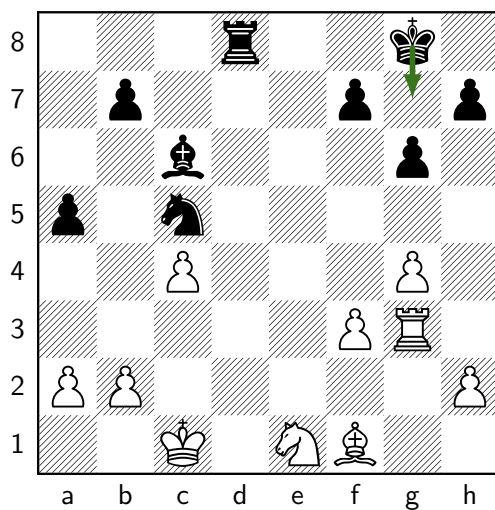


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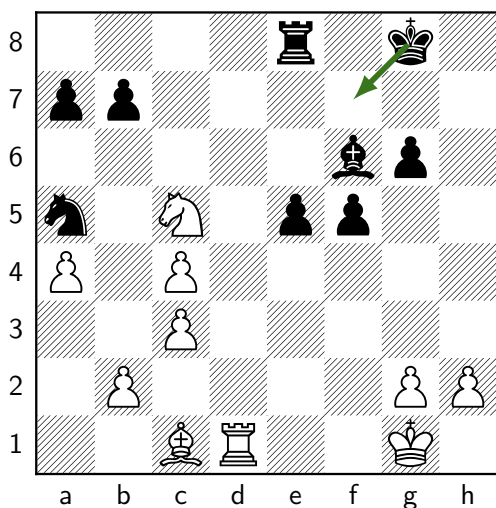
Figure 22: **Move 31, Black: Bg2**. (a) Our position; Melanie plays Bg2 (b7g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#622` (ply 52) [structural distance $d=41$, QAP obj=4]: the master played Rc6; under μ our Bb2 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Pb6 as their Nd2, our Ra8 as their Pg2; transferred move b7g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3425` (ply 54) [structural distance $d=44$, QAP obj=4]: the master played Ba5; under μ our Bb2 plays the role of that game’s Bb6; cross-type roles: our pc4 as their Rd1, our Nf6 as their Pf2; transferred move b7g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3568` (ply 46) [structural distance $d=47$, QAP obj=4]: the master played Rd6; under μ our Bb2 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our pf4 as their Nc3, our Nf6 as their Pg3; transferred move b7g2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bg2 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



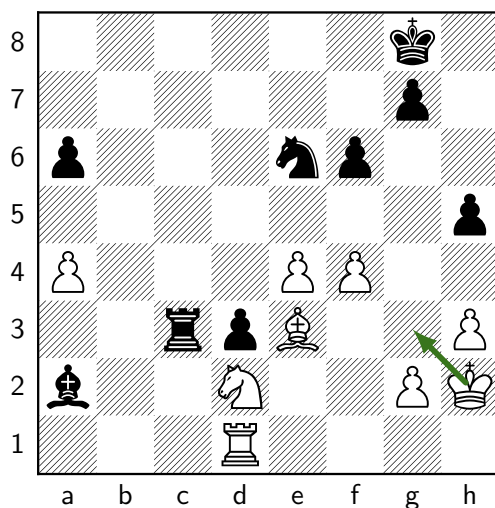
(a)



(b)



(c)



(d)

Figure 23: **Move 32, Black: Kg6 (g7g6)**. (a) Our position; Melanie plays Kg6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1069` (ply 51) [structural distance $d=36$, QAP obj=22]: the master played Kg7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Bg2 as their Rd8, our Ra8 as their Pf7; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#620` (ply 55) [structural distance $d=41$, QAP obj=19]: the master played Kf7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Pd7 as their Re8, our ra1 as their Bf6; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1846` (ply 72) [structural distance $d=36$, QAP obj=19]: the master played Kg3; under μ our Kg2 plays the role of that game’s Kh2; cross-type roles: our Pd7 as their Rd1, our na3 as their ba2; transferred move g7g6 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kg6 (summed QAP objective 122, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

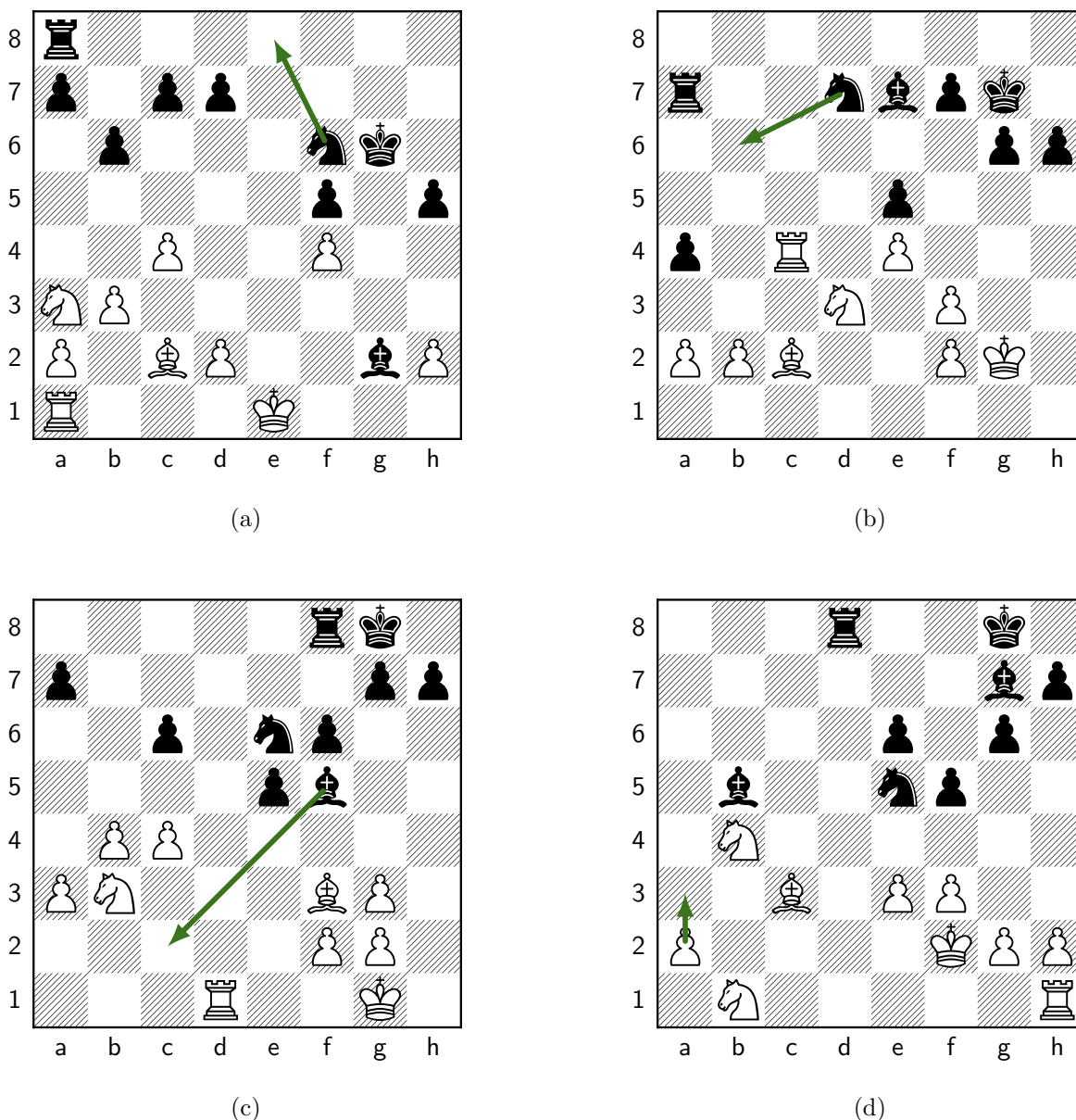
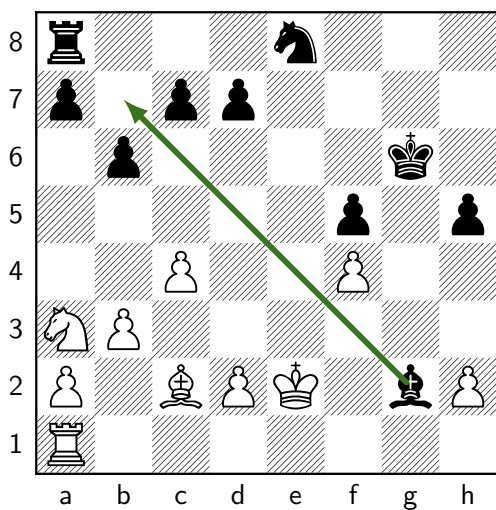
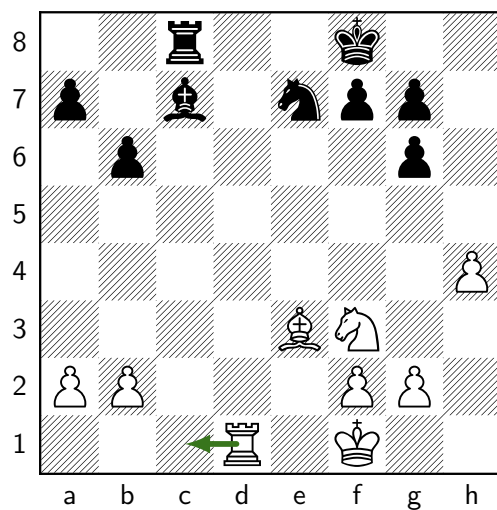


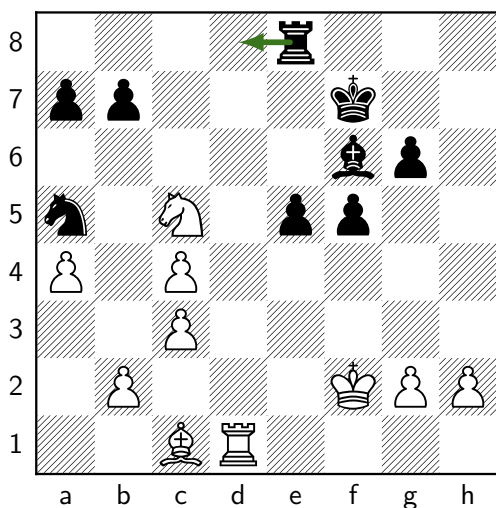
Figure 24: **Move 33, Black: Ne8 (f6e8)**. (a) Our position; Melanie plays Ne8 (f6e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2134` (ply 67) [structural distance $d=48$, QAP obj=26]: the master played Nb6; under μ our Nf3 plays the role of that game’s Nd2; cross-type roles: our na3 as their Be7, our bc2 as their rc4; transferred move f6e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3374` (ply 53) [structural distance $d=45$, QAP obj=21]: the master played Bc2; under μ our Nf3 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our pa2 as their Rf8, our Ra8 as their Pa7; transferred move f6e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3864` (ply 66) [structural distance $d=46$, QAP obj=9]: the master played a3; under μ our Nf3 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Pd7 as their Nb1, our Pb6 as their Bc3; transferred move f6e8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ne8 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



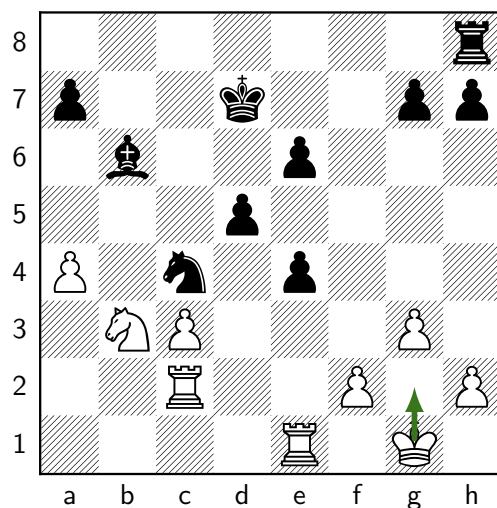
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(b)

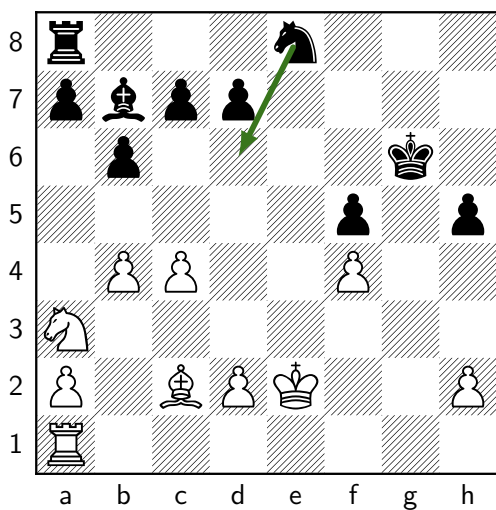


(c)

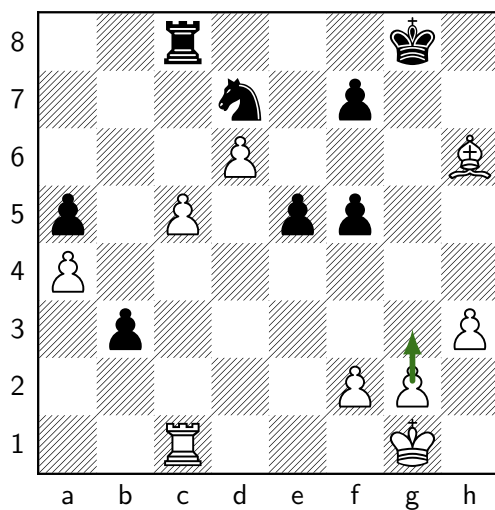


(d)

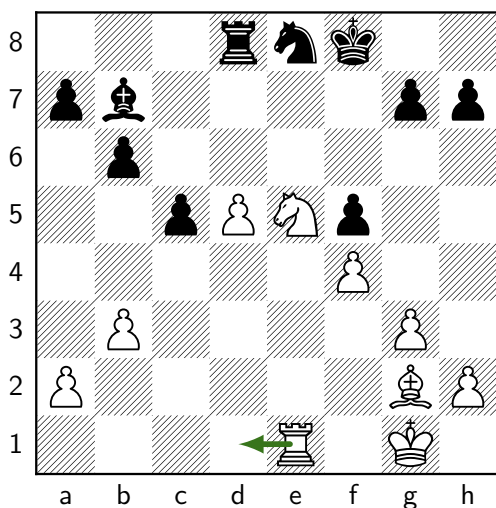
Figure 25: **Move 34, Black: Bb7**. (a) Our position; Melanie plays Bb7 (g2b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3831` (ply 50) [structural distance $d=48$, QAP obj=22]: the master played Rc1; under μ our Bg7 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pa2, our Ne8 as their Pb2; transferred move g2b7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#620` (ply 57) [structural distance $d=42$, QAP obj=19]: the master played Rd8; under μ our Bg7 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our na3 as their Pa7, our Pc7 as their Bf6; transferred move g2b7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1414` (ply 56) [structural distance $d=47$, QAP obj=46]: the master played Kg2; under μ our Kg3 plays the role of that game’s Kg1; cross-type roles: our bc2 as their Re1, our pc4 as their Rc2; transferred move g6h6. Synthesis: 3 retrieved analogs transferred to Bb7 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



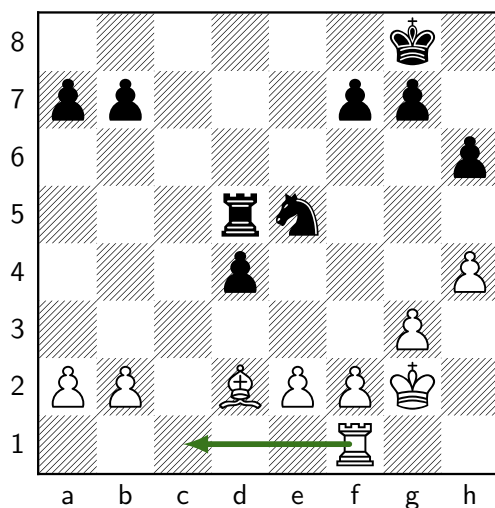
(a)



(b)

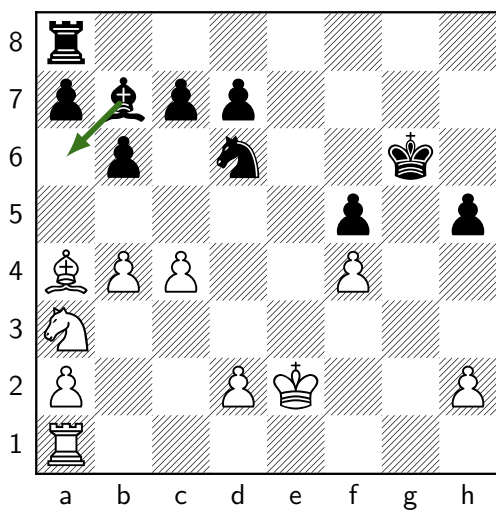


(c)

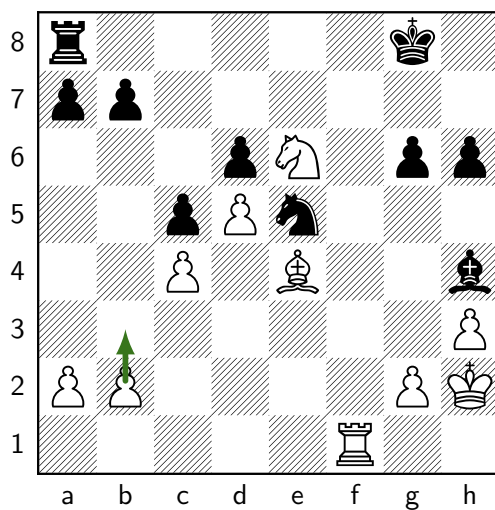


(d)

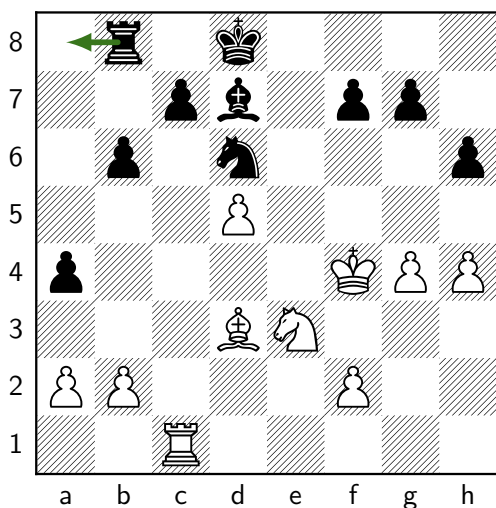
Figure 26: **Move 35, Black: Nd6.** (a) Our position; Melanie plays Nd6 (e8d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2058` (ply 62) [structural distance $d=41$, QAP obj=41]: the master played g3; under μ our *Ne1* plays the role of that game’s *Pg2* (a **cross-type** correspondence); cross-type roles: our *na3* as their *Rc1*, our *Ra8* as their *Pf2*; transferred move e8d6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1975` (ply 54) [structural distance $d=47$, QAP obj=35]: the master played Rd1; under μ our *Ne1* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *na3* as their *Bg2*, our *Bb7* as their *Ne5*; transferred move e8d6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1300` (ply 52) [structural distance $d=47$, QAP obj=16]: the master played Rc1; under μ our *Ne1* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *ra1* as their *Pf2*, our *Bb7* as their *Pg3*; transferred move e8d6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nd6 (summed QAP objective 121, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



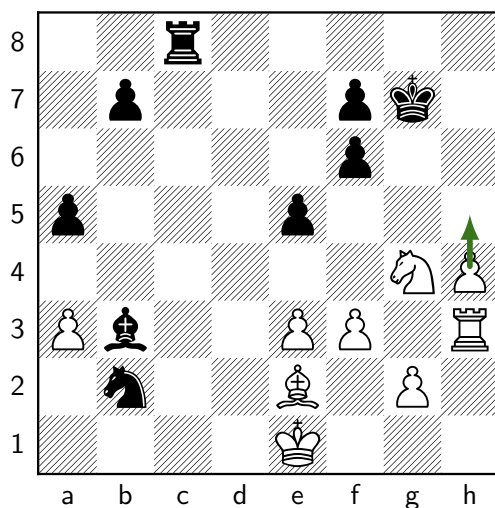
(a)



(b)



(c)



(d)

Figure 27: **Move 36, Black: Ba6**. (a) Our position; Melanie plays Ba6 (b7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2279` (ply 60) [structural distance $d=41$, QAP obj=45]: the master played b3; under μ our *Bb2* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *pa2* as their *bh4*, our *na3* as their *ph6*; transferred move b7a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#751` (ply 61) [structural distance $d=38$, QAP obj=21]: the master played Ra8; under μ our *Bb2* plays the role of that game’s *Rb1* (a **cross-type** correspondence); cross-type roles: our *Nd6* as their *Bd7*, our *pb4* as their *Nd6*; transferred move b7a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#380` (ply 56) [structural distance $d=33$, QAP obj=39]: the master played h5; under μ our *Ph4* plays the role of that game’s *Ph4*; cross-type roles: our *pf4* as their *nb2*, our *Pf5* as their *Be2*; transferred move h5h4. Synthesis: 2 retrieved analogs transferred to Ba6 (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

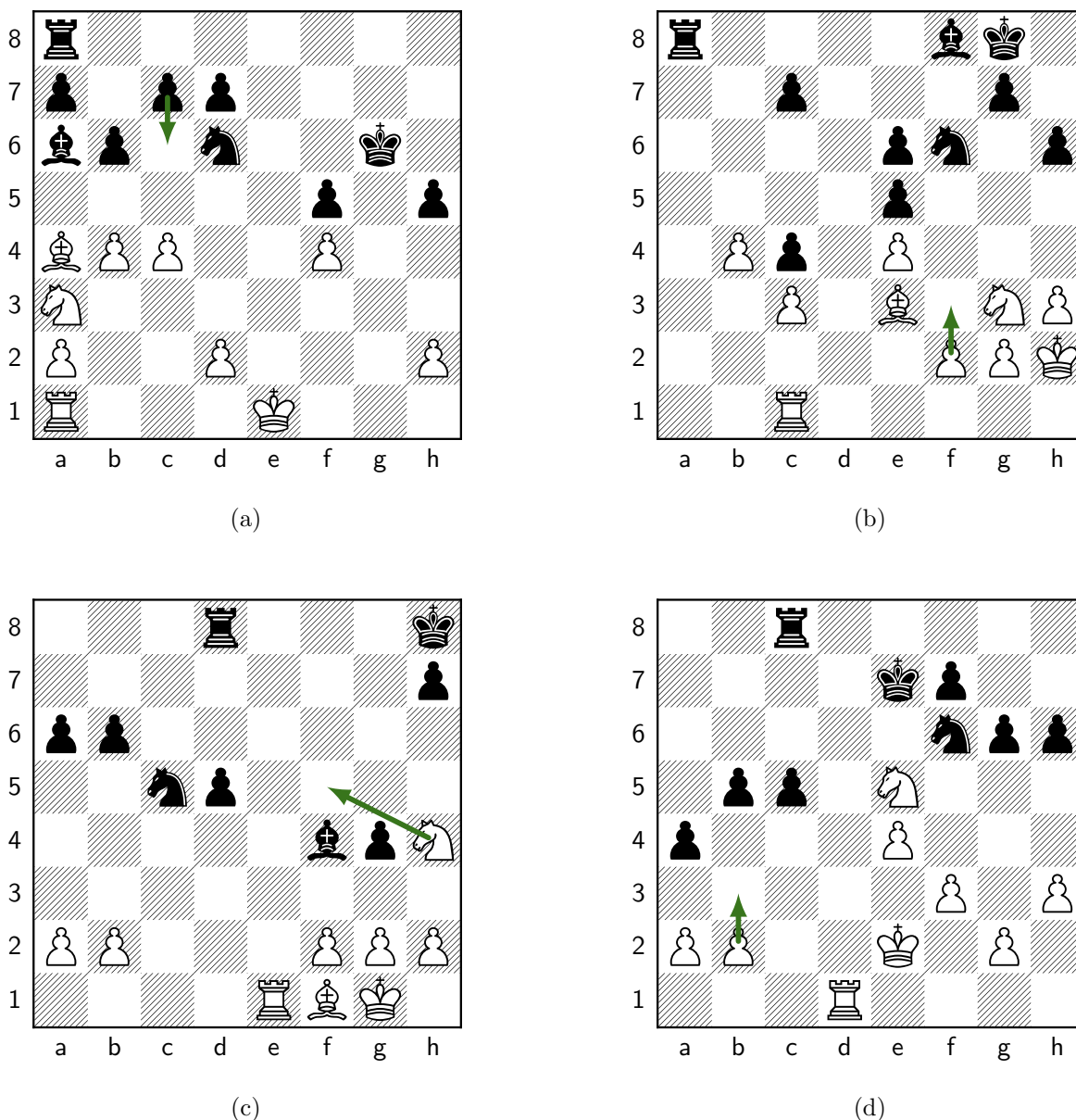
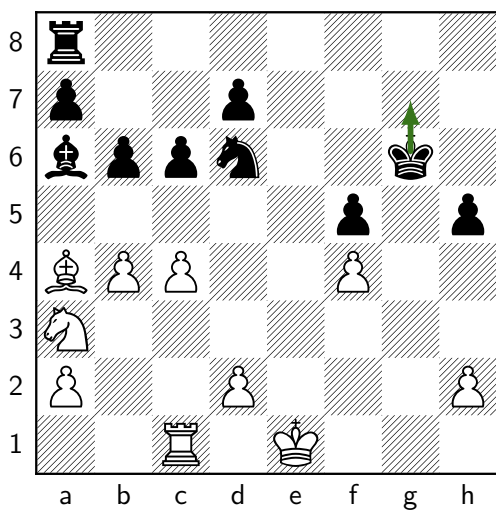
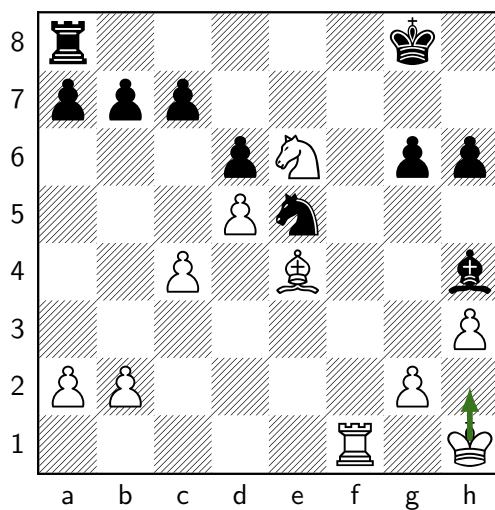


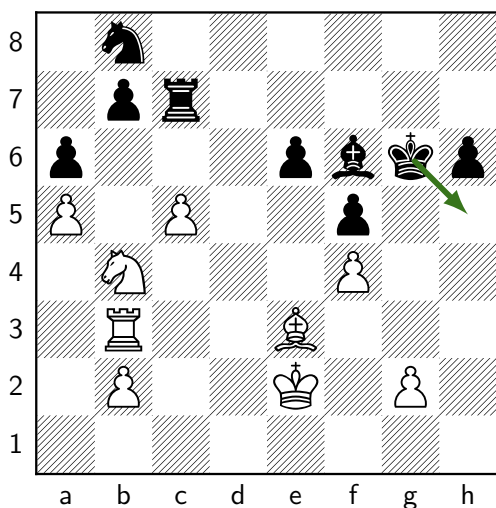
Figure 28: **Move 37, Black: c6**. (a) Our position; Melanie plays c6 (c7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1245` (ply 68) [structural distance $d=51$, QAP obj=39]: the master played f3; under μ our *Pc2* plays the role of that game’s *Pf2*; cross-type roles: our *Pf5* as their *Rc1*, our *Nd6* as their *Be3*; transferred move c7c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2421` (ply 62) [structural distance $d=43$, QAP obj=32]: the master played Nf5; under μ our *Pc2* plays the role of that game’s *Nh4* (a **cross-type** correspondence); cross-type roles: our *Pa7* as their *Re1*, our *Ra8* as their *Pb2*; transferred move c7c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3361` (ply 60) [structural distance $d=51$, QAP obj=52]: the master played b3; under μ our *Ba3* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *pd2* as their *Rd1*, our *na3* as their *pa4*; transferred move a6b7. Synthesis: 2 retrieved analogs transferred to c6 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



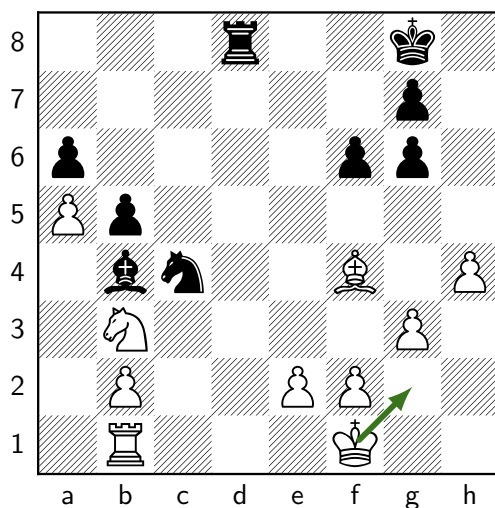
(a)



(b)

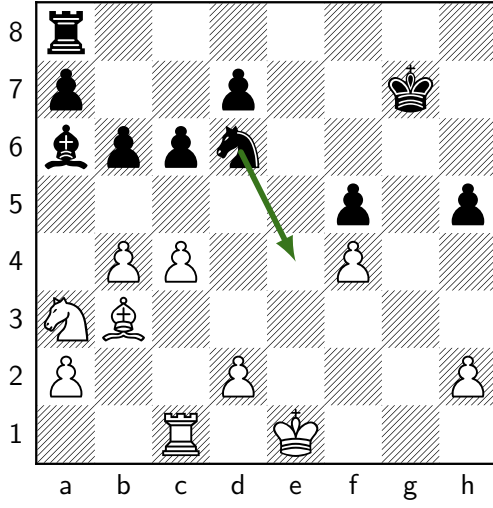


(c)

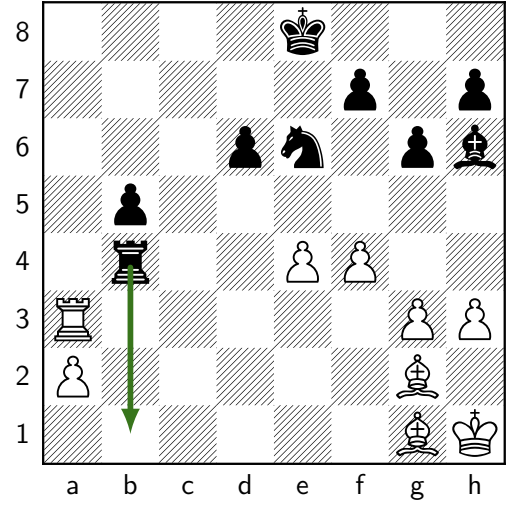


(d)

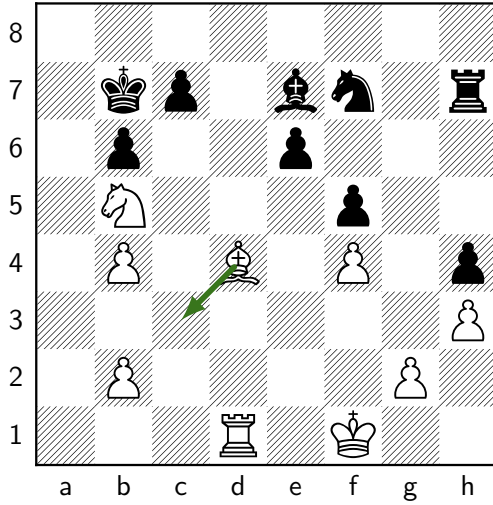
Figure 29: **Move 38, Black: Kg7**. (a) Our position; Melanie plays Kg7 (**g6g7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2279` (ply 58) [structural distance $d=46$, QAP obj=53]: the master played Kh2; under μ our Kg3 plays the role of that game’s Kh1; cross-type roles: our pd2 as their Rf1, our Ra8 as their Pa2; transferred move **g6g7 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#2737` (ply 83) [structural distance $d=47$, QAP obj=49]: the master played Kh5; under μ our Kg3 plays the role of that game’s Kg3; cross-type roles: our Pd7 as their Nb8, our Ba6 as their Rc7; transferred move **g6g7 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#830` (ply 68) [structural distance $d=49$, QAP obj=37]: the master played Kg2; under μ our Kg3 plays the role of that game’s Kf1; cross-type roles: our Pa7 as their Rb1, our Ra8 as their pf6; transferred move **g6g7 (endorses this move)**. Synthesis: 8 retrieved analogs transferred to Kg7 (summed QAP objective 239, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



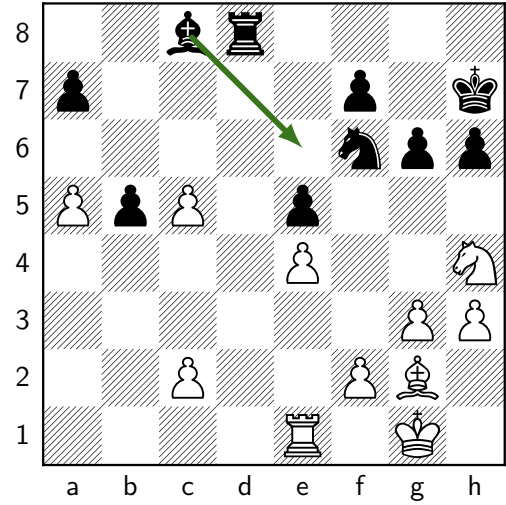
(a)



(b)

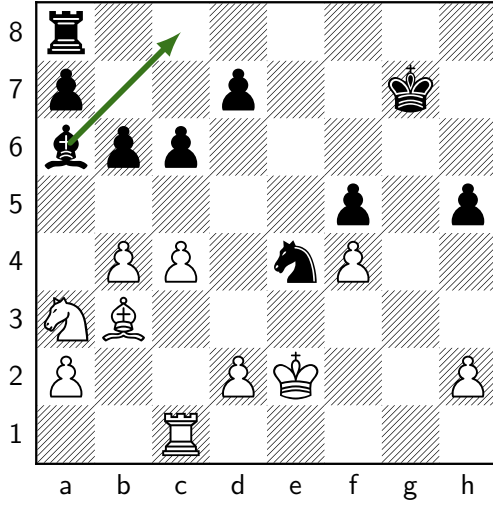


(c)

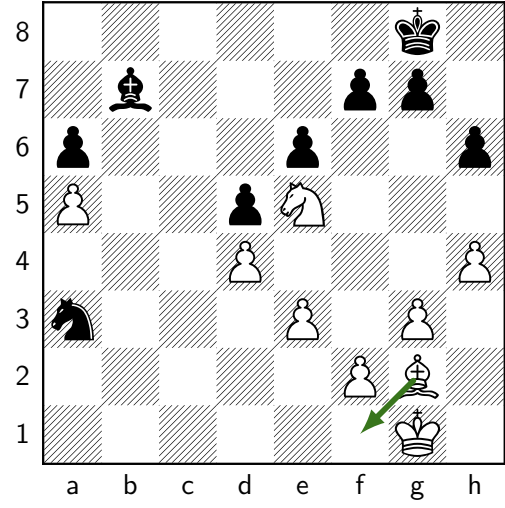


(d)

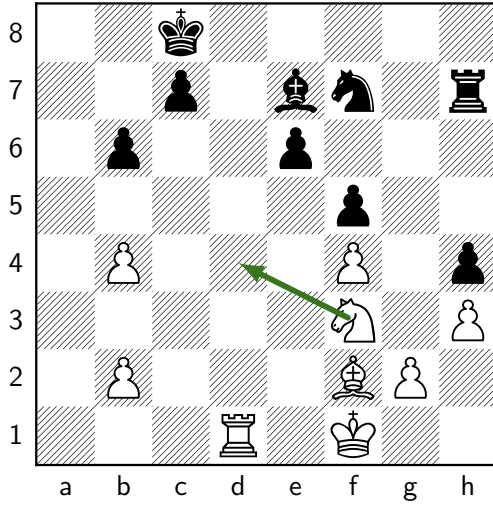
Figure 30: **Move 39, Black: Ne4**. (a) Our position; Melanie plays Ne4 (d6e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2608` (ply 75) [structural distance $d=46$, QAP obj=51]: the master played Rb1; under μ our Nd3 plays the role of that game’s Rb5 (a **cross-type** correspondence); cross-type roles: our na3 as their ra3, our bb3 as their pg3; transferred move d6e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2589` (ply 80) [structural distance $d=51$, QAP obj=41]: the master played Bc3; under μ our Nd3 plays the role of that game’s Bd4 (a **cross-type** correspondence); cross-type roles: our Ba6 as their Nb5, our na3 as their pf5; transferred move d6e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#355` (ply 61) [structural distance $d=33$, QAP obj=38]: the master played Be6; under μ our Nd3 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pc6 as their Rd8, our Ba6 as their Nf6; transferred move d6e4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Ne4 (summed QAP objective 213, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



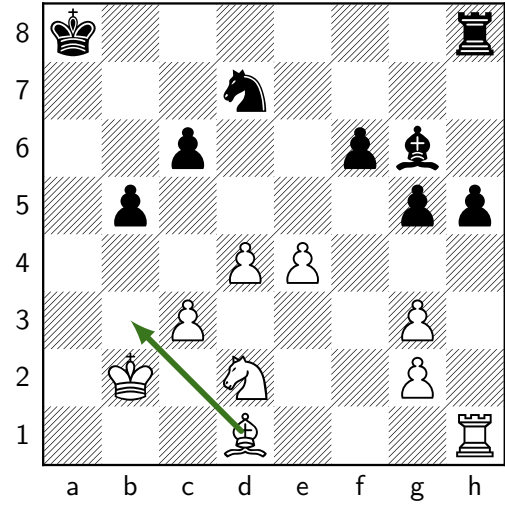
(a)



(b)

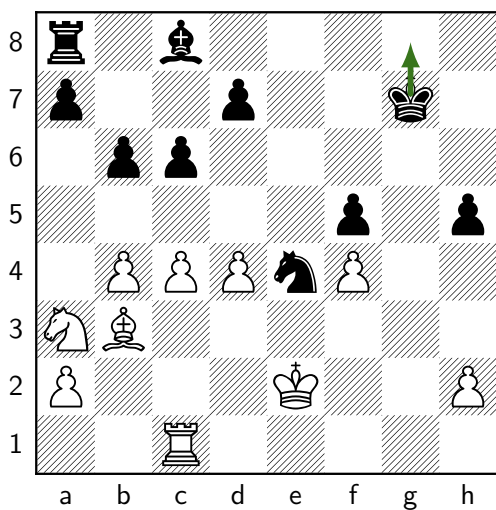


(c)

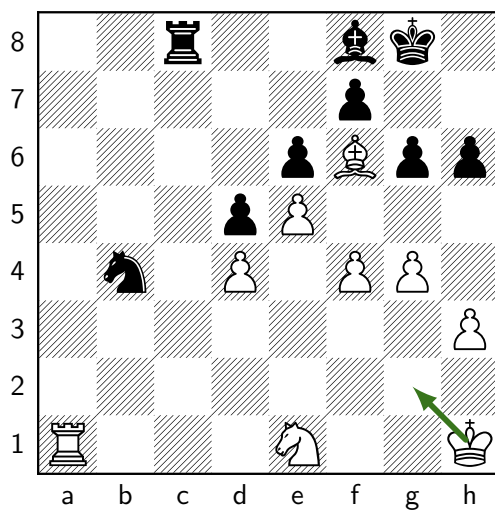


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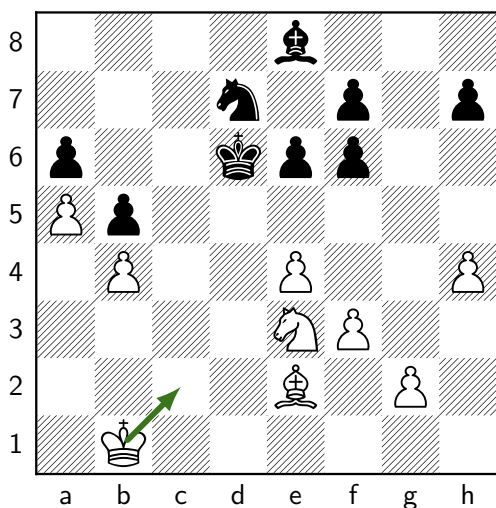
Figure 31: **Move 40, Black: Bc8.** (a) Our position; Melanie plays Bc8 (a6c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0002#2919` (ply 68) [structural distance $d=50$, QAP obj=55]: the master played Bf1; under μ our Ba3 plays the role of that game’s Bg2; cross-type roles: our pf4 as their na3, our Ra8 as their Pg3; transferred move a6c8 (**endorses this move**). (c) Retrieved from `ccr1.top_p0003#2589` (ply 70) [structural distance $d=50$, QAP obj=55]: the master played Nd4; under μ our Ba3 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our pf4 as their Rd1, our Ra8 as their Pb2; transferred move a6c8 (**endorses this move**). (d) Retrieved from `ccr1.top_p0000#2657` (ply 72) [structural distance $d=47$, QAP obj=50]: the master played Bb3; under μ our Ba3 plays the role of that game’s Bd1; cross-type roles: our Ba6 as their Rh1, our rc1 as their Nd2; transferred move a6c8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Bc8 (summed QAP objective 312, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



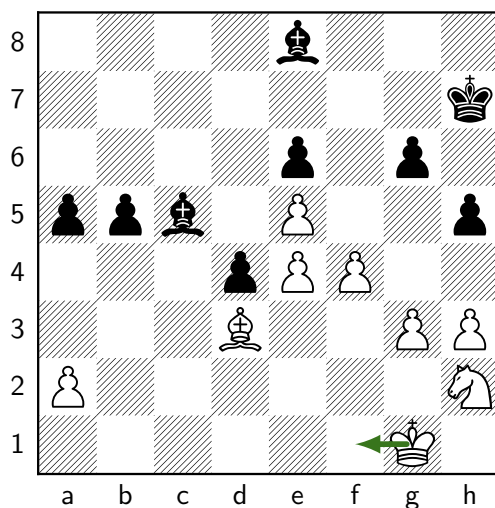
(a)



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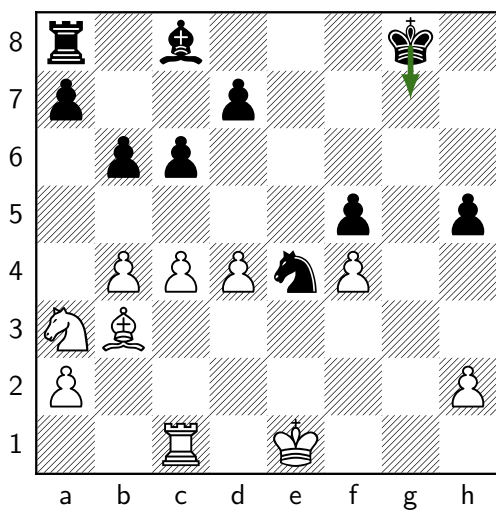


(c)

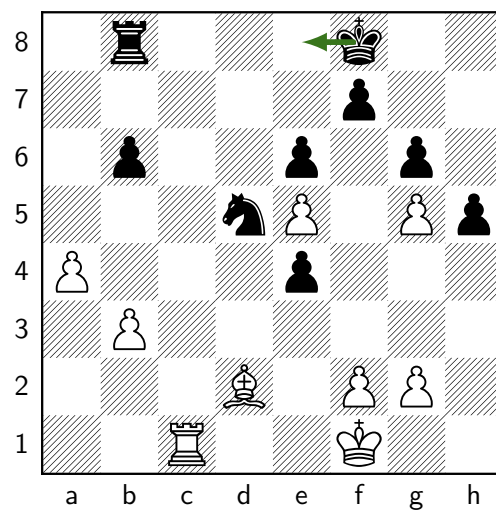


(d)

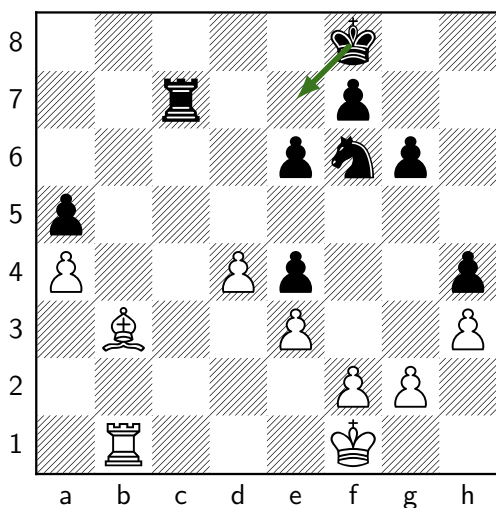
Figure 32: **Move 41, Black: Kg8**. (a) Our position; Melanie plays Kg8 (g7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0000#26` (ply 82) [structural distance $d=54$, QAP obj=6]: the master played Kg2; under μ our Kg2 plays the role of that game’s Kh1; cross-type roles: our pb4 as their Ra1, our pc4 as their Ne1; transferred move g7g8 (**endorses this move**). (c) Retrieved from `ccr1_top.p0003#1159` (ply 70) [structural distance $d=47$, QAP obj=6]: the master played Kc2; under μ our Kg2 plays the role of that game’s Kb1; cross-type roles: our Ne4 as their Be2, our Ra8 as their Pg2; transferred move g7g8 (**endorses this move**). (d) Retrieved from `ccr1_top.p0000#1027` (ply 74) [structural distance $d=49$, QAP obj=5]: the master played Kf1; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Pf5 as their Nh2, our na3 as their Bd3; transferred move g7g8 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kg8 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



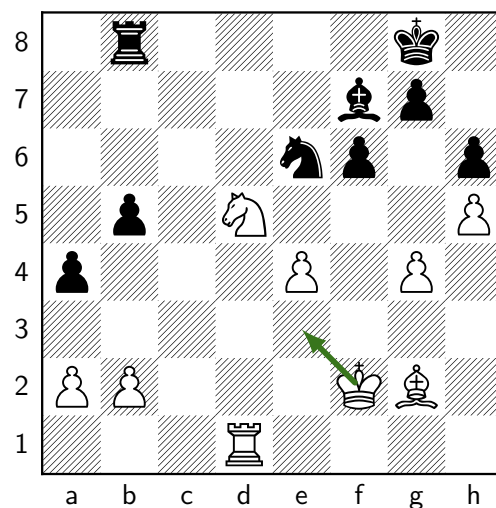
(a)



(b)

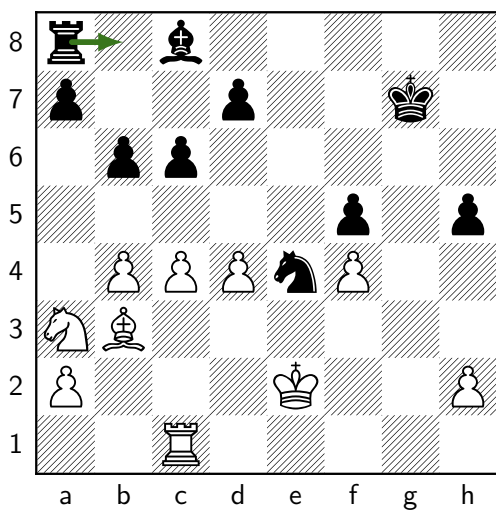


(c)

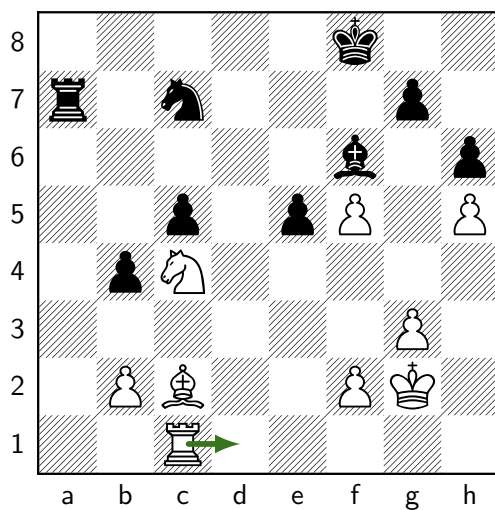


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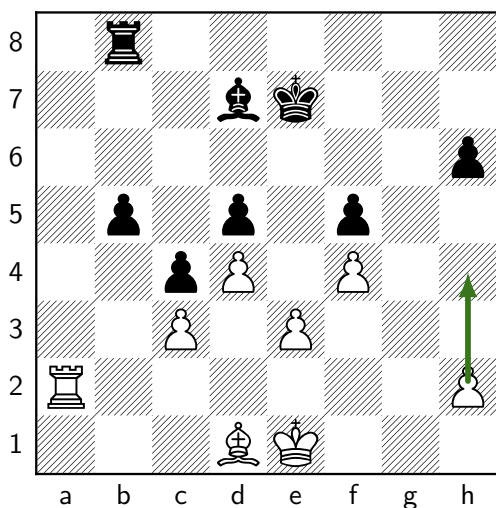
Figure 33: **Move 42, Black: Kg7.** (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1036` (ply 83) [structural distance $d=47$, QAP obj=5]: the master played Ke8; under μ our Kg1 plays the role of that game’s Kf1; cross-type roles: our Pd7 as their Rb8, our na3 as their Pb6; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3634` (ply 75) [structural distance $d=56$, QAP obj=5]: the master played Ke7; under μ our Kg1 plays the role of that game’s Kf1; cross-type roles: our pb4 as their Rc7, our Ra8 as their Pg6; transferred move g8g7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3521` (ply 68) [structural distance $d=52$, QAP obj=5]: the master played Ke3; under μ our Kg1 plays the role of that game’s Kf2; cross-type roles: our pf4 as their Bg2, our na3 as their Pg4; transferred move g8g7 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kg7 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



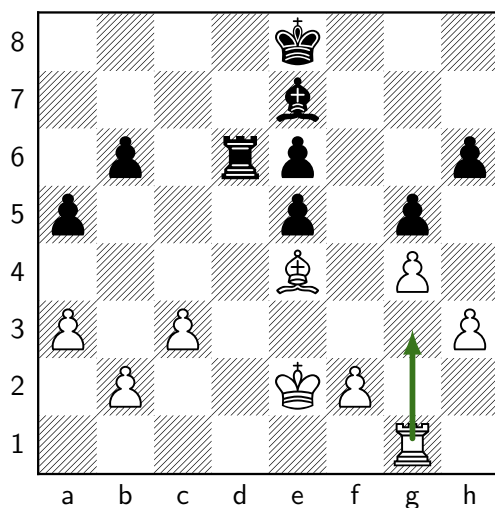
(a)



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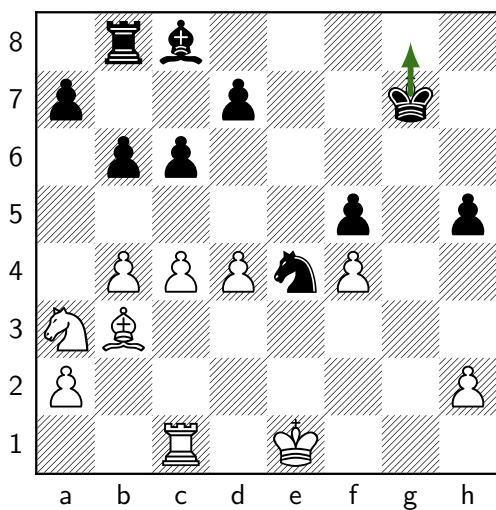


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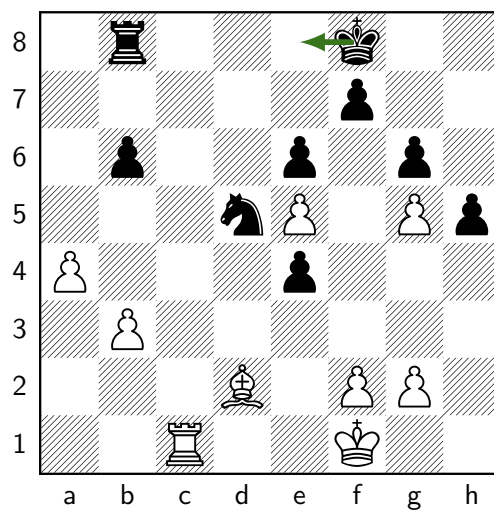


(d)

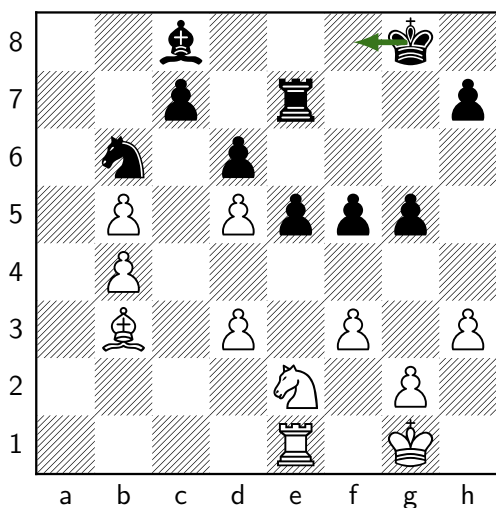
Figure 34: **Move 43, Black: Rb8 (a8b8)**. (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3786` (ply 72) [structural distance $d=52$, QAP obj=6]: the master played Rd1; under μ our *Ra1* plays the role of that game’s *Rc1*; cross-type roles: our *Ne4* as their *pb4*, our *pf4* as their *Nc4*; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3424` (ply 78) [structural distance $d=53$, QAP obj=4]: the master played h4; under μ our *Ra1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *Ne4* as their *Bd1*, our *Pb6* as their *Ra2*; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2324` (ply 82) [structural distance $d=40$, QAP obj=4]: the master played Rg3; under μ our *Ra1* plays the role of that game’s *Rg1*; cross-type roles: our *Bc8* as their *Pf2*, our *Pb6* as their *Be4*; transferred move a8b8 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Rb8 (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



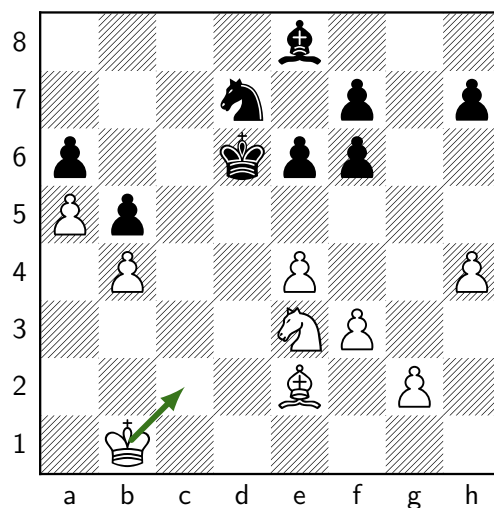
(a)



(b)

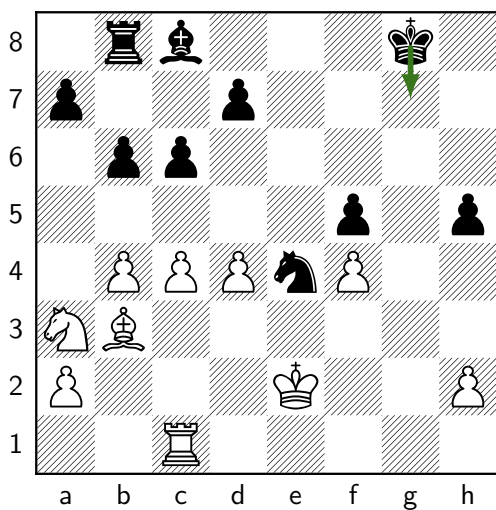


(c)

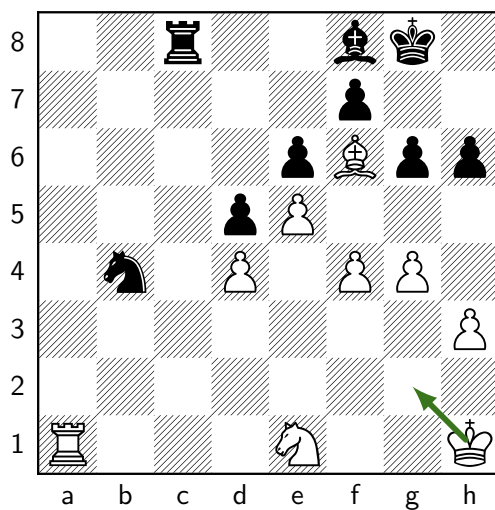


(d)

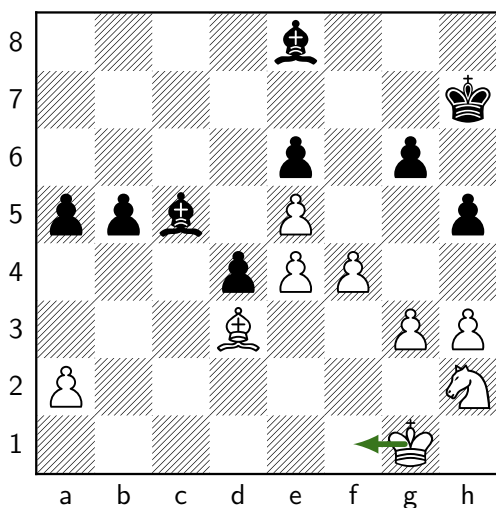
Figure 35: **Move 44, Black: Kg8.** (a) Our position; Melanie plays Kg8 (g7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1036` (ply 83) [structural distance $d=48$, QAP obj=5]: the master played Ke8; under μ our Kg2 plays the role of that game’s Kf1; cross-type roles: our Pa7 as their Rb8, our bb3 as their Pb6; transferred move g7g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1017` (ply 81) [structural distance $d=56$, QAP obj=5]: the master played Kf8; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our na3 as their Bc8, our bb3 as their Re7; transferred move g7g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1159` (ply 70) [structural distance $d=55$, QAP obj=4]: the master played Kc2; under μ our Kg2 plays the role of that game’s Kb1; cross-type roles: our Pd7 as their Be2, our pb4 as their Ne3; transferred move g7g8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kg8 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



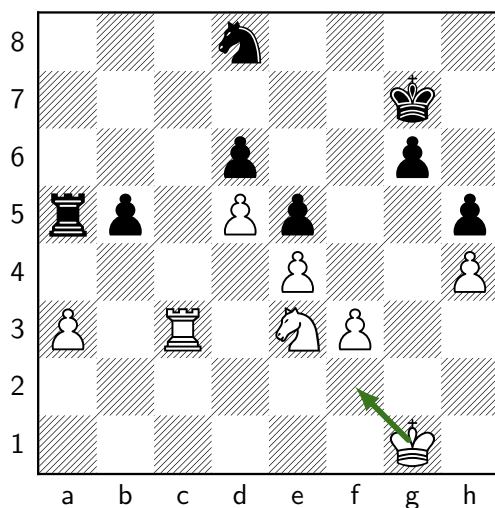
(a)



(b)

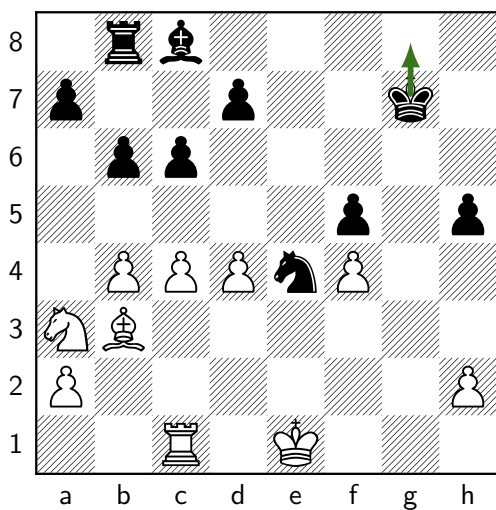


(c)

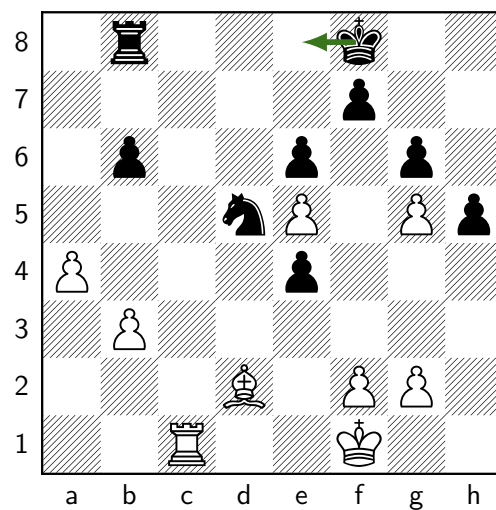


(d)

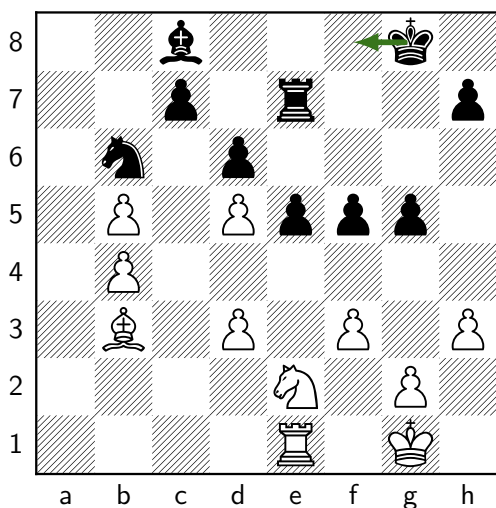
Figure 36: **Move 45, Black: Kg7**. (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#26` (ply 82) [structural distance $d=54$, QAP obj=6]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kh1; cross-type roles: our Pc6 as their Ra1, our Pd7 as their Ne1; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1027` (ply 74) [structural distance $d=49$, QAP obj=5]: the master played Kf1; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our pb4 as their Nh2, our Pd7 as their Bd3; transferred move g8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3486` (ply 72) [structural distance $d=56$, QAP obj=5]: the master played Kf2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our bb3 as their Rc3, our pc4 as their Ne3; transferred move g8g7 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kg7 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



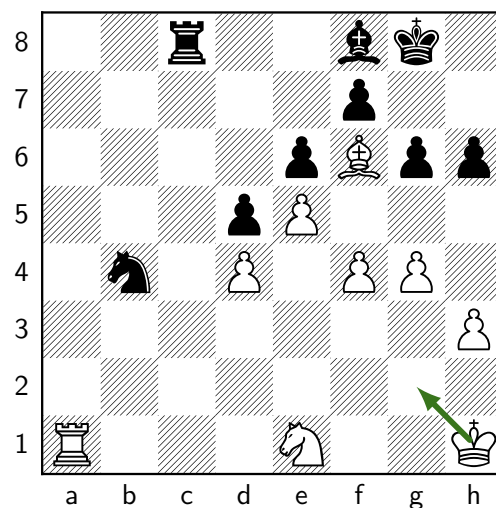
(a)



(b)



(c)



(d)

Figure 37: **Move 46, Black: Kg8**. (a) Our position; Melanie plays Kg8 (g7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1036` (ply 83) [structural distance $d=48$, QAP obj=5]: the master played Ke8; under μ our Kg2 plays the role of that game’s Kf1; cross-type roles: our Pa7 as their Rb8, our bb3 as their Pb6; transferred move g7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1017` (ply 81) [structural distance $d=56$, QAP obj=5]: the master played Kf8; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our na3 as their Bc8, our bb3 as their Re7; transferred move g7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#26` (ply 82) [structural distance $d=58$, QAP obj=5]: the master played Kg2; under μ our Kg2 plays the role of that game’s Kh1; cross-type roles: our Pf5 as their Ra1, our Bc8 as their Ne1; transferred move g7g8 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kg8 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.